When Leakage Changes the Conclusion: A Methodological Evaluation of Segmentation-Guided Burn Severity Classification

Ryan Altayeb [1], Abdulrahman Alraddadi [1], Mohannad Alrehaili [1]

1 Faculty of Computer and Information Systems, Islamic University of Madinah, Madinah, Saudi Arabia

Abstract

A burn photograph contains the wound and a great deal besides: unburned skin, dressings, bedding, the room. A classifier trained on whole images may read the setting rather than the injury. The obvious remedy is to segment the burn and grade only what remains, a design that is standard elsewhere in medical imaging but had not been tested for burns against the simpler alternative it is meant to improve on. We build a segmentation-guided pipeline — a YOLOv8x-seg localiser feeding a masked Swin classifier — and evaluate it against a plain classifier under one protocol on shared test data. Masking helps only when the mask is perfect, and then only slightly: pooled across architectures and seeds the benefit is $+0.55$ percentage points (95 percent confidence interval -0.37 to $+1.47$). Because that ablation uses ground-truth masks, it is an upper bound on what any segmenter could contribute. End to end, once the classifier is trained on the same kind of masks it is tested on, the two systems are statistically indistinguishable on both test sets. Segmentation-guided classification attains parity, not advantage, and runs two models to do so. On an independent clinical database it does not transfer at all, assigning the mildest grade to half of a set that contains no mild burns. The methodological result is the one we would emphasise: **leakage here did not merely inflate a score, it changed the conclusion.** Splitting the augmented files at random left 80.5 percent of the masked test set sharing a source photograph with training, while the separately partitioned unmasked set leaked none. Because the leak rate differed between the two conditions being compared, it did not simply raise a number. It manufactured a design recommendation — 2.07 percentage points in favour of masking where a corrected split gives 0.55 — consistently across 20 of 21 architectures, which is the pattern an author reads as robustness. Six further evaluation choices, each conventional in this literature, each changed an answer we obtained. Three seeds gave an internal interval three times narrower than ten support while missing the external effect entirely. Choosing the comparison arm on test rather than validation accounts for 71 percent of its margin, so we report no internal effect estimate at all. None of these failures is specific to burns. Code, weights, per-image predictions and a script recomputing every published number: `https://github.com/Just-Ryan/burn-severity-benchmark`.

Keywords

Burn severity assessment, deep learning, data leakage, external validation, image segmentation, reproducibility

Article informations

Corresponding author: altayeb.ray@gmail.com

1. Introduction

Burns are common and serious injuries. About 11 million people need medical attention for burns each year, and the World Health Organization estimates that burns cause roughly 180,000 deaths annually, most of them in low- and middle-income countries where specialist care is scarce (World Health Organization, 2023; Peck, 2011). Survivors often face long hospital stays, scarring, and disability (Brusselaers et al., 2010). Deciding how to treat a burn depends first on assessing its depth, which separates injuries that heal with conservative care from those that need surgery (Jackson, 1953). Burns are graded as first degree (epidermis only), second degree (partial thickness, into the dermis), or third degree (full thickness) (Heimbach et al., 1984).

Depth assessment is difficult. Even experienced burn surgeons assess burn depth accurately only 64 to 76 percent of the time (Jaskille et al., 2009), and it is affected by lighting, wound evolution, blistering, and skin pigmentation (Monstrey et al., 2008). Instrumented alternatives such as infrared thermography reach high accuracy but are expensive and rarely available (Dang et al., 2021). An automated, reproducible assessment aid could therefore be useful, particularly where specialists are absent.

Deep learning has reached expert-level performance on several medical imaging tasks (Esteva et al., 2017; Litjens et al., 2017), and it has been applied to burns for two coupled subproblems, segmenting the burn region and classifying its severity (Boissin et al., 2023). Progress is real, but the burn literature has three recurring gaps. First, most studies train and report a single architecture, which makes it hard to know whether a reported number reflects the method or the particular network. Second, few studies report confidence intervals, significance tests, or evaluation on data from a source other than the training set. Third, and rarely examined, is data hygiene: whether the split procedure itself inflates the reported numbers. These gaps matter because high in-distribution accuracy has repeatedly failed to predict external performance in medical imaging, and because the evaluation choices that produce such numbers are themselves under active scrutiny. Leakage has recently been shown to inflate prediction performance across a range of mechanisms in neuroimaging (Rosenblatt et al., 2024), and community guidance on metric selection for biomedical image analysis was substantially revised in 2024 (Maier-Hein et al., 2024). A recent five-year systematic review of the burn literature finds accuracies of 80 to 98 percent reported on single datasets. It judges six of fourteen studies to carry a high risk of bias, faults the reliance on public datasets that do not represent the target population, and reports dataset details as frequently unstated (Fu, 2026). The evidence that region-of-interest masking helps at all is likewise mixed (Sourget et al., 2025). Both considerations shape how we evaluate the design below: in-distribution accuracy is not by itself evidence of clinical usefulness, and we treat leakage-controlled external evaluation as the standard the pipeline has to meet.

The idea we set out to test. A burn photograph contains the wound and a great deal besides: unburned skin, dressings, bedding, clothing, a clinician's gloved hand, the room. A classifier trained on whole images is free to use any of it. In a web-sourced collection that freedom is dangerous, because background correlates with severity for reasons that have nothing to do with tissue: severe burns are photographed in hospitals, minor ones at home. A model can score well by reading the setting rather than the injury, and such a model will fail the moment the setting changes. The obvious remedy is to remove everything that is not the wound. Segment the burn, multiply the mask into the image so the background is black, and grade only what remains. The classifier can then attend to the properties clinicians actually use, the colour, texture and blanching of the injured tissue, because nothing else survives.

That reasoning is sound enough that the design is standard elsewhere in medical imaging, where dual-stage frameworks localise a region of interest before classifying it (Bruno et al., 2025; Al-masni et al., 2020). For burns it had not been evaluated. The published burn work either classifies whole images or segments them, and where the two are combined the combination is not tested against the simpler alternative it is supposed to improve on. Whether isolating the burn actually helps a burn-severity classifier was, when we began, an open question with an assumed answer.

We built the system to find out: a YOLOv8x-seg localiser feeding a masked Swin transformer classifier, served through a Flutter and Flask mobile application. We then built a benchmark to test it honestly, and the honest test is the substance of this paper.

What that test produced is the claim this paper is organised around. **In this study the evaluation protocol, not the architecture, determined the conclusion.** Under a file-level split the experiment says masking helps, across 20 of 21 architectures. Under a source-grouped split, on the same images and the same models, it says masking does not help. No network was changed between those two answers.

We should therefore be explicit about what this paper is and is not. **We propose no new architecture, no new loss and no new training procedure, and we claim no state of the art.** Every model here is an off-the-shelf ImageNet-pretrained backbone or a stock YOLOv8x-seg. What we contribute is a measurement of how much the evaluation protocol itself decides the answer in segmentation-guided medical image classification, carried out on a case where we happened to be the ones misled. **Our contributions are three.**

First, the method and its evaluation. We specify a segmentation-guided burn-severity pipeline and evaluate it against the alternative it is meant to beat, a plain classifier on unmasked images, under one protocol on shared test data. We give the pipeline its best available configuration rather than a convenient one. The masking ablation uses ground-truth masks, so it bounds from above what any segmenter could contribute. The classifier arm is the strongest architecture in its condition. And we ask whether the pipeline's deficit comes from mask quality or from a mismatch: it trains on ground-truth masks but meets predicted ones at inference.

Second, a benchmark built to be trusted rather than to flatter. A source-grouped split of 1,370 photographs audited with perceptual hashing, 11 ImageNet-pretrained architectures under four input conditions, a ten-seed replication rather than the customary three, Wilson and source-clustered intervals, paired tests kept separate by estimand, and two independent external sets, one of them clinical. Every number we report is recomputed from released raw data by a script that ships with the paper.

Third, a measurement of how much the evaluation protocol itself decides the answer. Building that benchmark obliged us to correct choices that are routine in this literature, and the corrections were not cosmetic. Splitting an augmented dataset at the file level manufactured a masking benefit of 2.07 percentage points that falls to 0.55 (95 percent confidence interval -0.37 to $+1.47$) once the split is grouped by source photograph. Three training seeds, the usual number, gave an internal interval three times narrower than ten seeds support while missing the external effect entirely; of all 120 three-seed subsets of ten runs, only 15 detect it. The internal comparison holds on plain accuracy and not on balanced accuracy. And selecting the comparison arm on test rather than validation accounts for 71 percent of the internal margin. We report these because a reader cannot judge our answer without them, and because they generalise past burns: they are properties of the protocol, not of this dataset.

The remainder of the paper is organised as follows. Section 2 places the design in the literature. Section 3 describes the dataset, the source-grouped split and its leakage controls, the pipeline, the benchmark, and the evaluation protocol. Section 4 reports segmentation quality, whether masking helps the classifier, whether the pipeline helps end to end, external validation, a skin-tone probe, efficiency, and a sensitivity analysis of the evaluation protocol itself. Section 5 interprets these results and states the limitations, and Section 7 concludes.

2. Related work

Automated burn assessment and the high single-dataset pattern. Computer-aided burn assessment predates deep learning: early systems on the Seville BIP_US database combined hand-crafted colour and texture features with classical classifiers to separate burn depths (Acha et al., 2005). Deep learning has since become dominant and is typically applied to two coupled subproblems, delineating the burn region and grading its severity (Boissin et al., 2023). Recent work spans convolutional classifiers (Suha and Sanam, 2022), attention-augmented residual networks (Yadav et al., 2023), YOLO-family detectors adapted to burn severity (Ferdinand et al., 2023; Yıldız et al., 2024), and transformer-based frameworks that perform segmentation and severity assessment jointly (Nadeem et al., 2026). Several report high single-dataset accuracy, for example 95.6 percent with a transfer-learning classifier (Suha and Sanam, 2022). A recent five-year systematic review reports single-dataset accuracies of 80 to 98 percent while judging six of fourteen studies to be at high risk of bias, largely because the public datasets used do not represent the target population (Fu, 2026). Which severity class is hardest is not settled in this literature: Nadeem et al. (2026) report second degree as among their best-performing classes and attribute their weakest classes to under-representation, whereas second degree is consistently the hardest class in our own results.

Comparative studies on a single split. Comparative burn studies exist (Rahman et al., 2024; Chang et al., 2023), but they evaluate several backbones on a single split and rarely report confidence intervals, significance tests, or external evaluation. Without these, apparent improvements may not be reliable and out-of-distribution behaviour is unknown. Our work targets exactly these gaps: we compare approaches under one fixed protocol, report Wilson confidence intervals, apply paired significance tests, and add explicit external tests.

Segment-then-classify as an established design, and whether masking helps. Cascading a segmentation stage into a downstream classifier is well established in medical imaging. Dual-stage frameworks localise a region of interest and then classify it in breast ultrasound (Bruno et al., 2025) and in dermoscopy (Al-masni et al., 2020), and the same cascade appears in burns for wound measurement and surface-area estimation (Chang et al., 2021). This grounds our two-stage design as a standard template, not a novelty. The usual motivation is that masking to the lesion suppresses background shortcuts and focuses the classifier on relevant tissue. The evidence that this helps is mixed: models trained on full images can match or exceed models restricted to the region of interest because networks exploit background and spurious cues, so masking is not a universally beneficial preprocessing step (Sourget et al., 2025). This is consistent with our central negative result.

Mobile and point-of-care deployment and its validation gap. Because our contribution is a deployed system, the most relevant comparators are fielded tools. The most clinically validated is the DeepView multispectral device, whose artificial-intelligence ensemble predicted burn healing potential with 100 percent specificity in a feasibility study (Thatcher et al., 2023), albeit with dedicated multispectral hardware rather than a consumer camera. Smartphone-only wound tools are more common but less validated: a deep-learning pressure-injury staging app reported 63.2 percent accuracy on its 144-image validation set (Lau et al., 2022), and a review of mobile ulcer-segmentation apps concluded that most disclose neither their training data nor adequate clinical validation (Griffa et al., 2024). Against this backdrop, a consumer-phone burn system such as ours, like the mobile burn app of Yıldız et al. (2024), is valuable precisely when it is reported honestly, including where it underperforms.

Data leakage, external validation, and the central theme. Leakage, information about the test set contaminating training, is a long-recognised cause of inflated, irreproducible results (Kaufman et al., 2012), and it has been identified as a systemic driver of the reproducibility crisis across machine-learning-based science (Kapoor and Narayanan, 2023). In medical imaging, a systematic review that screened 2,212 studies and adjudicated 62 found none of them of potential clinical use, with leakage, biased curation, and inadequate reporting among the leading causes (Roberts et al., 2021). Community responses include consensus reporting standards such as the REFORMS checklist (Kapoor et al., 2024). A large body of evidence shows that in-distribution accuracy is a poor predictor of external accuracy. A pneumonia network that scored an internal area under the curve of 0.931 fell to 0.815 at an external hospital and was found to exploit site-specific artifacts (Zech et al., 2018). A systematic review of externally validated radiology models found external performance routinely lower and frequently unreported (Yu et al., 2022), and reliable in-distribution magnetic-resonance models degraded sharply on out-of-distribution cohorts (Mårtensson et al., 2020). A concrete and easily reproduced form of leakage in image work is splitting after augmentation, so that augmented copies of one source image land in both folds. Quantified case studies of this kind exist, and they report inflations larger than ours. Splitting brain MRI at the slice rather than the subject level inflated accuracy by between 29 and 55 percent across four cohorts (Yagis et al., 2021), and an equivalent failure in retinal and breast OCT inflated accuracy by 5 to 30 percent (Tampu et al., 2022). The picture is not uniform: in connectome-based prediction, some forms of leakage inflate performance substantially while others have minor or even negative effects (Rosenblatt et al., 2024). What these cases share is that leakage moves the absolute score of a single model. Closer to our case, confound removal, applied as a corrective, has itself been shown to introduce leakage into one arm of a comparison and so to manufacture an apparent benefit for that arm (Hamdan et al., 2023). Our study places burn classification within this pattern, and treats leakage-controlled external evaluation as the standard against which the design is judged.

How many training runs a conclusion needs. That a benchmark result depends on more than the reported point estimate is established. Comparing two methods from single runs produces high rates of both false positives and false negatives. The variance contributed by perturbing the data split exceeds that contributed by weight initialisation alone (Bouthillier et al., 2021) — an ordering our own results reproduce, since our split procedure moved the masking answer further than our seed did. Even where the overall spread across seeds is modest, individual outlier seeds that perform markedly above or below the average are easy to find (Picard, 2021). In medical imaging specifically, evaluation noise has been shown to exceed the gap between the winning model and the best 10 percent of entries in five of eight challenges examined (Varoquaux and Cheplygina, 2022). Biomedical image analysis rankings are unstable to methodological choices such as the metric variant or the removal of a single test case (Maier-Hein et al., 2018). What we add is not the observation that seeds matter, which belongs to this literature. It is a measured instance in which the conventional three-run protocol changed which of two conclusions the evidence supported, in both directions at once (Section 4.6), together with the full distribution of answers the same data would have yielded (Section 4.9).

3. Materials and methods

3.1 Dataset

The primary dataset was the publicly available "skin burn wound classification" dataset on Roboflow Universe (version 31) (Binus, 2023), released under a Creative Commons Attribution 4.0 licence and exported in COCO segmentation format. It comprises **1,370 unique burn photographs** in three severity classes; the export augments these into about 3,424 annotated files with distinct file names (about 2.5 copies per source). Table 1 summarises the dataset, the leak-free split, and the two external sets. We map the numeric source labels to first, second, and third degree. Class shares are approximately 42 percent first degree, 31 percent second, and 27 percent third. The labels are community-provided and were not independently verified by clinicians.

From the split we derive a segmentation dataset (polygon annotations, either a single "burn" class or three severity classes) and a classification dataset in four input forms. These are unmasked full images; masked images, in which the ground-truth polygon annotation is rasterised to a binary mask and multiplied into the image so that non-burn pixels are zero; images cropped to the annotation bounding box; and masked and cropped. We emphasise that the masked form uses ground-truth annotations rather than predicted masks, so any benchmark result on it is an oracle upper bound, a distinction we return to when interpreting the pipeline.

3.2 Source-grouped split and leakage control

The naive split follows common practice and partitions the augmented image files at random. Because the source dataset augments each photograph into several copies with distinct file names, a file-level split scatters copies of one photograph across training and test. In the masked classification set built this way, re-split 70/15/15 after flattening, **417 of its 518 test images (80.5 percent) shared a source photograph with training**. The unmasked set was never re-split: it retained the collection's own 90/6/4

Table 1: Dataset, leak-free split, and external sets. The naive file-level split (documented only to explain the leakage artifact) shared a source photograph between test and training for 80.5 percent of masked test images and none of the unmasked test images.

Item	Value
Primary set (Roboflow "skin burn wound classification", version 31, CC BY 4.0)	
Unique source photographs	1,370
Augmented files	$\approx$ 3,424
Split (by source, stratified by class, seed 42)	70 / 15 / 15
Validation / test (deduplicated to one image per source)	206 / 205
Internal test distribution (first / second / third)	86 / 63 / 56
Naive-split source overlap into training (masked / unmasked)	80.5% / 0%
External sets	
BIP_US clinical database (Seville; second-vs-third probe)	94
Clean external set (second Roboflow collection, pHash-verified)	319
distribution (first / second / third)	139 / 157 / 23
images removed as perceptual-hash identical to training	118
sources contributing a removed image	56 / 199 (28%)
sources eliminated entirely	24 / 199

partition, whose 143-image test fold leaked none. The two arms of that original comparison therefore differed in split ratio and in test-set size as well as in leak rate, which we state because it makes the confound larger than a single-split description would suggest, not smaller. This asymmetry, together with memorisation of near-duplicate training images, manufactured an apparent masking benefit of 3.06 percentage points across 20 of 21 architectures. That figure and the leak-free figure below come from different architecture pools, so we also report the matched contrast: restricted to the eight architectures common to both analyses, the naive split gives $+2.07$ percentage points (7 of 8 positive) against $+0.55$ on the source-grouped split. The matched contrast, $2.07 \rightarrow 0.55$, is the smaller of the two and is the one we rely on; we give the 21-architecture figure because it is what the original experiment produced. We report the leaky result only to document the artifact, not as performance, and we never use it as a headline result.

The correction is a source-grouped split. We partition the 1,370 source photographs 70/15/15, stratified by class and seeded at 42, keep all augmented copies only in the training fold, and deduplicate validation and test to one image per source (206 and 205 images). All downstream leak-free numbers use this split. The internal test set thus contains **205** images, distributed 86 first degree, 63 second degree, and 56 third degree.

Source grouping removes leakage between augmented copies of the same file, but it cannot remove what the file names do not encode. Auditing the corrected split with the same perceptual-hash procedure we later apply to the external set (Section 3.6), we find **2 of the 205 test images (1.0 percent) are perceptually identical to a training image carrying a different source identifier**. That is, the same photograph was contributed to the collection twice under different names. One of the two pairs is additionally annotated with conflicting labels in the two folds. We report this rather than suppress it, for three reasons. First, the effect is bounded, though not as harmlessly as we first wrote. Every reported accuracy moves by at most 1.0 percentage point, which is within the cross-seed standard deviation. But the paired ten-seed internal difference is itself only $+2.63$ points. If both images were counted as errors for the classifier and correct for the pipeline on every seed, that difference falls to $+0.68$ with a 95 percent interval of -1.66 to $+3.03$ and $p = 0.53$. That is the worst case, and an implausible one, but a $4/205$ swing rather than a $2/205$ one. Even the milder scenario in which only the classifier loses the two images leaves $+1.66$, interval -0.69 to $+4.00$. Neither excludes zero. The external comparison, which carries the paper's argument, is unaffected because these two images are internal. We record this because an earlier draft of this paragraph claimed the duplicates "reverse no comparison in this paper", and for the internal paired comparison under its worst case that was not true. Second, it is a concrete instance of the paper's own argument, namely that leakage is easy to introduce and hard to see, and that a single grouping heuristic is not a proof of independence. Third, it yields a practical recommendation we now follow ourselves: *grouping by identifier should be verified with a content-based check*, because identifiers describe provenance only as well as the upstream collection was curated.

3.3 The segmentation-guided pipeline

The deployed system processes an image in two stages (Figure 1). A single-class YOLOv8x-seg localiser predicts a burn mask; the mask is binarised and multiplied into the image so that the background is black; the masked image is passed to a Swin transformer classifier (Swin-Tiny), which outputs the severity grade. The system is served through a Flask backend and a Flutter mobile front-end. The mobile demonstration currently ships a Swin-Small classifier, whereas all pipeline numbers reported in this paper use the benchmarked Swin-Tiny weights. We stress that the application is a research and demonstration artifact, not a clinically validated device, and Section 4.8 shows that it does not transfer to independent clinical images.

3.4 Benchmark design

We run two benchmarks that answer two distinct questions.

Benchmark 1 asks whether focusing the classifier on the burn helps. We train 11 ImageNet-pretrained architectures, a representative subset of the 21 used in the naive-split experiment chosen to span convolutional and transformer families within the compute budget, on the leak-free split: ConvNeXt-Tiny, ConvNeXt-Large, DenseNet201, EfficientNet-B0, MobileNetV3-Large, ResNet50, Swin-Tiny, Swin-Small, Swin-Base, Swin-Large, and ViT-Base (Liu et al., 2022; Huang et al., 2017; Tan and Le, 2019; Howard et al., 2019; He et al., 2016; Liu et al., 2021; Dosovitskiy et al., 2021). Coverage of the four input conditions of Section 3.1 is not uniform, and we state it explicitly. All 11 architectures were trained under the unmasked and masked conditions, whereas the two cropping conditions were run on 8 of the 11, omitting ConvNeXt-Tiny, Swin-Base, and Swin-Large for compute reasons. Because the masked condition uses ground-truth masks, its result is an oracle upper bound on any masking benefit that a real segmenter could achieve.

Benchmark 2 asks whether segmentation-first helps end to end, through a head-to-head on the shared 205-image internal test set and on the external sets. The two arms are not symmetric, and we state the asymmetry rather than claim fairness. The standalone classifier is evaluated in its best Benchmark 1 condition, ConvNeXt-Large on unmasked images, whereas the pipeline arm is fixed to the deployed configuration, Swin-Tiny on masked images, because the deployed system is what this paper is about. This favours the classifier arm, and the comparison should be read as the best available plain classifier against the deployed pipeline configuration, not as the best of each family. The conclusion does not rest on that choice alone: Benchmark 1 independently bounds the *average* masking benefit at about $+1.5$ percentage points even with ground-truth masks. That is a bound on the pooled mean across architectures, not on any single one, and Section 4.3 shows the per-architecture effect is not stable; it therefore constrains, without excluding, the possibility that some untested masked configuration does better. The robust pipeline runs the localiser at a low confidence threshold of 0.05, masks the image, and classifies it with Swin-Tiny, falling back to the full image when the localiser returns no region, which matches the deployed application. That threshold was inherited from the deployed system rather than tuned, and we checked after the fact whether it flatters the results. Sweeping the localiser confidence from 0.001 to 0.5 on the 206-image *validation* fold selects 0.005, which yields 77.07 percent on test against the 78.05 percent obtained at 0.05, and the test-optimal value is a third setting again (0.001, 79.51 percent). The reported operating point is therefore neither the validation optimum nor the test optimum, and the whole sweep spans about two percentage points until the threshold rises past 0.25 and detections start being lost. The all-in-one YOLOv8-seg model is evaluated at the same low confidence threshold of 0.05 for fairness. We report accuracy and, on the imbalanced external sets, balanced accuracy.

For robustness, the Benchmark 2 classifiers were trained and evaluated over seeds 0, 1, and 2 and are reported as mean and sample standard deviation (denominator $n-1$). The Benchmark 1 masking ablation was confirmed over seeds 0, 1, and 42 for the masked-versus-unmasked comparison on 8 of the architectures; the cropping conditions were evaluated at seed 42 only. These are separate experiments, run at different times, and we report each seed set as it stands rather than merging them.

3.5 Training protocol

Classifiers used ImageNet-pretrained backbones (Deng et al., 2009) from the timm library (Wightman, 2019), each with the classification head resized to three outputs and trained under one fixed recipe. That recipe was the AdamW optimiser, class-weighted cross-entropy using inverse class frequencies, a cosine schedule with early stopping, mixed precision, and input resized to 224×224. Training-time augmentation (horizontal flip, colour jitter, small rotation) was enabled and evaluation-time augmentation disabled, with an identical transform for every architecture so that differences reflect the network rather than the setup.

Both YOLOv8x-seg models (the single-class localiser and the three-class standalone model) used the same configuration: 100 epochs at 640×640 input, batch size 8, the AdamW optimiser at an initial learning rate of 10^{-3}, seed 42, and early stopping with patience 20. Both were initialised from the public `yolov8x-seg.pt` weights (Jocher et al., 2023).

An important asymmetry between the two stages, which we discovered late and report in full. The source-grouped split of Section 3.2 was constructed for the *classification* datasets. The segmentation models were

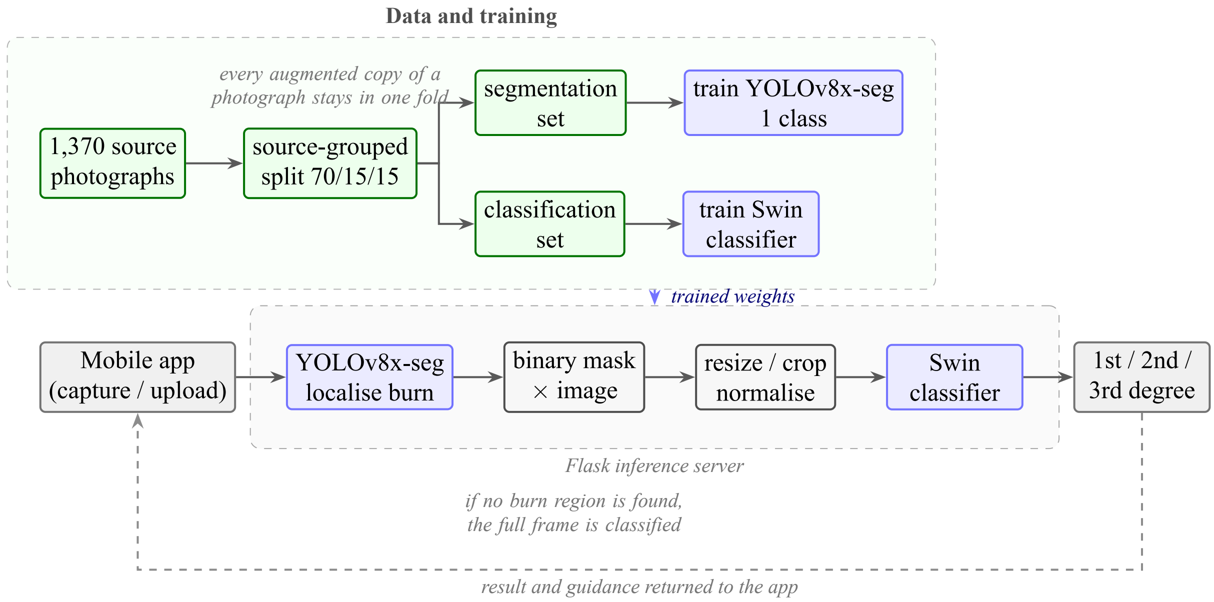


Figure 1: The two-stage system, from data to prediction. *Above:* the 1,370 source photographs are partitioned once, by source photograph rather than by file, and that one partition defines the folds for both models. This is the step that Section 4.11 shows we originally applied to the classifiers alone. *Below:* at inference a mobile application sends an image to a Flask backend, where the YOLOv8x-seg localiser predicts a burn mask, the mask is multiplied into the image so that the background is black, and the Swin classifier grades what remains as first, second, or third degree. If the localiser finds nothing, the full frame is classified instead, which is the behaviour we evaluate throughout. The system is a research and demonstration artifact, not a clinically validated device.

trained earlier, on the collection's original distribution-supplied split, and we did not rebuild them when we rebuilt the classifier split. The two splits do not agree. Auditing them against each other, **179 of the 205 internal classification test images (87.3 percent) fall in the segmentation models' training or validation folds**: 145 (70.7 percent) in training and 34 in validation, leaving only 26 images that those models had never seen. Section 4.11 reports what this does and does not affect. In short, every internal number that passes through a YOLO model is optimistic; the segmentation quality figures, the entire Benchmark 1 masking ablation, the oracle-mask results and every external number are unaffected, and we verified rather than assumed each of those.

The Benchmark 1 masking ablation was run on a Kaggle NVIDIA P100 (PyTorch 2.5.1, CUDA 12.1) (Paszke et al., 2019). The Benchmark 2 segmentation training, the head-to-head classifier retraining, the external evaluation, and the efficiency measurements were run on a Google Colab NVIDIA A100 (PyTorch 2.11, CUDA 12.8). The retrained classifier accuracies are about two percentage points lower than the earlier seed-42 Kaggle run. Hardware, library version, recipe and seed set all differ between the two, so we cannot attribute that gap to any one of them and do not try; we report it so that the two sets of numbers are not mistaken for replicates of each other.

3.6 Evaluation protocol and statistics

For a like-for-like comparison, the three-class standalone YOLOv8-seg detections are reduced to a per-image majority-degree label, directly comparable to the classifier and the pipeline on the same 205 images. Images for which the localiser returns no region are never dropped from the denominator. Under the strict pipeline they are counted as errors; under the robust pipeline they fall back to the full frame and are scored on that, which is what the deployed application does. To make the effect of this choice transparent, we report both a strict pipeline (default confidence 0.25, no-detection counted as an error) and the robust pipeline (confidence 0.05 with full-image fallback) as an ablation.

We report accuracy with Wilson 95 percent confidence intervals (Wilson, 1927) and compare systems on the same images with the paired exact McNemar test (McNemar, 1947). Two distinct questions require two distinct tests, and we keep them separate throughout. Whether one system beats another *on a particular test set* is an image-level question, answered by McNemar on paired predictions. Whether one system beats another *on average across training runs* is a run-level question, answered by a paired test over per-seed accuracies. Overlapping error bars are not used as evidence of absence, since the comparisons are paired.

The masking effect was tested with the Wilcoxon signed-rank test over architecture-seed pairs, and we report it as an effect size with a confidence interval rather than against a single detectability threshold. The difference between transformer and convolutional architectures was tested with a one-sided Mann-Whitney test (the directional hypothesis being that transformers benefit more); the two-sided value is also reported in Section 4.3 for transparency. Following the honesty rule of this study, we prefer interval estimates to accept/reject statements wherever the data allow.

Two external protocols probe generalisation. The first uses the BIP_US clinical database of the Biomedical Image Processing Group at the University of Seville (94 photographs) (Acha et al., 2005; Rostami et al., 2021), which predates and is independent of the web-sourced training data. BIP_US labels depth as superficial dermal, deep dermal, or full thickness; because the mapping onto our three classes is not exact, we report both plausible mappings of deep dermal (to second degree and to third degree), giving a second-versus-third-degree probe with no first-degree cases. The second external set is a clean set of 319 images that we built from a second public Roboflow burn collection by removing every image that overlapped our training data. We screened for overlap with a perceptual-hash method: pHash and dHash computed with the `imagehash` library (Zauner, 2010), compared by Hamming distance against all primary training images and made horizontal-flip-aware. An image was treated as a duplicate when either hash matched a training image exactly, at Hamming distance 0, in any of the flip orientations tested; after removal we verified that no retained image lay within a Hamming distance of 10 of any training hash. The collection contained 1,371 images from 199 source photographs. The check removed 118 images that were perceptual-hash identical to training; those images came from 56 of the 199 sources (28 percent). Removal was at the image level, not the source level, so a source that contributed one contaminated image could retain its other copies, and it is the verified Hamming-distance-10 margin above, not source-level exclusion, that rules out residual contamination among the retained images. After also dropping the fourth-degree class and remapping the remainder to first, second, and third degree, 319 images remain, distributed 139, 157, and 23, drawn from **175 distinct source photographs** (mean 1.8 images per source, maximum 4). Of the original 199 sources, 24 are gone entirely. We state this arithmetic explicitly because 199 minus 56 is not 175, and a reader is right to check. Those 319 images are external to the training data, but they are not 319 statistically independent observations, so an interval computed on $N = 319$ is too narrow. Rather than only note this, we quantify it: alongside every external Wilson interval we report a cluster bootstrap that resamples the 175 *source photographs* with replacement (20,000 resamples, percentile method), which is the interval we rely on. Clustering widens the external accuracy intervals by roughly a fifth, from 9.6 to 11.7 percentage points for the classifier and from 10.0 to 11.9 for the pipeline. We retain the image-level point estimates for comparability with the internal test set. One consequence is worth stating because it is a defence rather than a concession: the ten-seed paired differences of Section 4.6 are computed over *training runs*, not over images, so source clustering does not widen those intervals. That defence is about the interval, not about generalisation: any claim beyond these particular 319 images still inherits the test set's limited source diversity. The perceptual-hash check is reported as a method, not an afterthought, because it reinforces the leakage and reproducibility theme.

3.7 Skin-tone probe

Because any imaging aid intended for clinical use should be checked for systematic dependence on skin tone, we added an exploratory probe. No ground-truth phototype labels exist for either dataset, so we estimated *apparent* skin tone from the images with the Individual Typology Angle, $\mathrm{ITA} = \arctan((L^* - 50)/b^*) \cdot 180/\pi$. ITA was computed in CIE $L^*a^*b^*$ under a D65 white point and binned into the six conventional bands of Chardon et al. (1991) (very light, light, intermediate, tan, brown, dark).

ITA must be measured on unburned skin, since a burn changes colour drastically. We therefore excluded the burn region from every image before sampling: on the internal set from the ground-truth mask, recovered from the masked and unmasked copies of the same photograph, and on the external set by rasterising the polygon annotations. Both exclusions were dilated by 6 pixels so that the burn margin could not contaminate the sample. Within the remaining region we selected skin pixels by the conjunction of the standard RGB (Kovac et al., 2003) and YCbCr skin rules and took the median ITA over those pixels. Images yielding fewer than 500 skin pixels were recorded as indeterminate and excluded rather than guessed at; this removed 11 of the 205 internal images and none of the external images, leaving 194 and 319.

We report accuracy per band with Wilson intervals, but we test the association on the *continuous* ITA with a Mann-Whitney test comparing the ITA of correctly and incorrectly classified images, so that no conclusion depends on where the band edges happen to fall.

The limits of this measurement are severe and we state them before reporting any result. ITA computed from uncontrolled photographs of unknown provenance is a proxy for apparent skin tone *in the image*, not a measurement of a patient's constitutive phototype: white balance, illumination, camera processing and perilesional erythema all shift it. The probe is exploratory and hypothesis-generating. It is not a fairness audit, and we do not present it as one.

4. Results

4.1 Segmentation quality

On the held-out test set the single-class localiser, which is the pipeline front-end, reached a mask mean average precision at intersection-over-union 0.50 (mask mAP50) of **0.726** and a mask mAP50-95 of 0.417. The three-class standalone model reached a mask mAP50 of **0.603** and a mask mAP50-95 of 0.349 (Table 2). The single-class model scores higher because it need not separate severity classes. Second degree is the hardest class to segment, consistent with the classifiers' weakest per-class performance on partial-thickness burns. Segmentation is therefore only moderate, a point that matters for the pipeline below.

4.2 Why the split matters: an augmentation-induced leakage artifact

Under the naive file-level split, masking appeared to raise accuracy for 20 of 21 architectures by a mean of 3.06 percentage points, and by 2.07 points across the eight architectures that also enter the pooled leak-free estimate (7 of 8 positive). The 2.07-point figure is the matched one the leak-free result should be compared against (Figure 2). This benefit does not survive scrutiny: it is a data-leakage artifact, driven by the 80.5 percent versus 0 percent source overlap between the masked and unmasked test sets (Table 1), where the masked accuracies were inflated by memorisation of near-duplicate training images.

What distinguishes this from the quantified leakage case studies of Section 2 is not its size, which is smaller than theirs, but its structure. In those studies leakage inflates the absolute score of a model. Here the leak rate was *confounded with the experimental condition under comparison*, 80.5 percent in the masked arm against none in the unmasked arm. Leakage therefore did not inflate a number but manufactured a comparative conclusion and, from it, a design recommendation, doing so consistently across 20 of 21 architectures. The nearest precedent we have found is confound-leakage, where a corrective preprocessing step introduces leakage into one arm of a comparison and so produces an apparent benefit for that arm (Hamdan et al., 2023). Our own mechanism is narrower than confound-leakage and we should not conflate the two. Masking introduces no leakage of itself, and what differed between our arms was the split protocol, one arm re-partitioned at file level and the other left on the collection's own partition (Section 3.2). Split-after-augmentation leakage is itself documented (Yagis et al., 2021; Tampu et al., 2022). What we have not found previously documented is the combination, an asymmetric split protocol that leaks into only one arm of a controlled comparison, with the per-arm leak rate measured on both sides. That combination, rather than any single ingredient, is what turned a null into a design recommendation. The transferable lesson is narrow and practical: when a leak rate differs between the arms being compared, consistency across architectures is evidence *for* the confound rather than against it, and is therefore not the reassurance it appears to be.

We document the leaky result only to explain the artifact. Every conclusion below rests on the source-grouped split, with the one exception recorded in Section 4.11, where the segmentation stage was found to have kept the original partition.

4.3 Does masking help the classifier?

On the leak-free split the masking benefit disappears (Table 3). For seed 42 alone, background masking raised accuracy by 1.5 percentage points overall, concentrated in the transformer architectures (3.1 percentage points, in all five of five) with roughly no change for the convolutional networks. That pattern did not replicate. Adding the fresh seeds 0 and 1 to seed 42, over the 24 architecture-seed pairs available in all three runs, collapses the effect to $+0.55$ **percentage points with a 95 percent confidence interval of** -0.37 **to** $+1.47$ (Wilcoxon signed-rank $p = 0.32$; 13 of 24 pairs improved). The interval is the informative quantity: it is consistent with no effect, and it rules out any benefit larger than about 1.5 percentage points. The apparent transformer advantage also failed to replicate (one-sided Mann-Whitney $p = 0.33$; two-sided $p = 0.66$), with transformers at $+0.77$ and convolutional networks at $+0.33$ percentage points pooled. Cropping to the burn produced essentially no change at seed 42, and masking with cropping likewise; the two cropping conditions were run on 8 of the 11 architectures and were not repeated across seeds, so we treat them as a weaker, single-seed check. We read this symmetrically: not that masking is exactly zero, but that any benefit it confers is smaller than the run-to-run variation of the training procedure itself, and that the seed-42 signal was noise. This matches the wider finding that region-of-interest masking is not reliably beneficial (Sourget et al., 2025).

4.4 Does the pipeline help end to end?

Table 4 reports the core comparison as our original protocol ran it, on three seeds. Section 4.6 replaces it with ten, and Figure 3 plots those ten runs; we give the three-seed figures here because they are what the earlier version of this work reported and because the difference between

Table 2: Segmentation quality on the held-out test set (Benchmark 2, YOLOv8x-seg). Both models used the identical configuration (100 epochs, 640×640, batch 8, AdamW at 10^{-3}, seed 42, patience 20). Second degree is the hardest class.

Model	Classes	Test mask mAP50	Test mask mAP50-95
Single-class localiser (pipeline front-end)	1	0.726	0.417
Standalone YOLOv8-seg	3	0.603	0.349

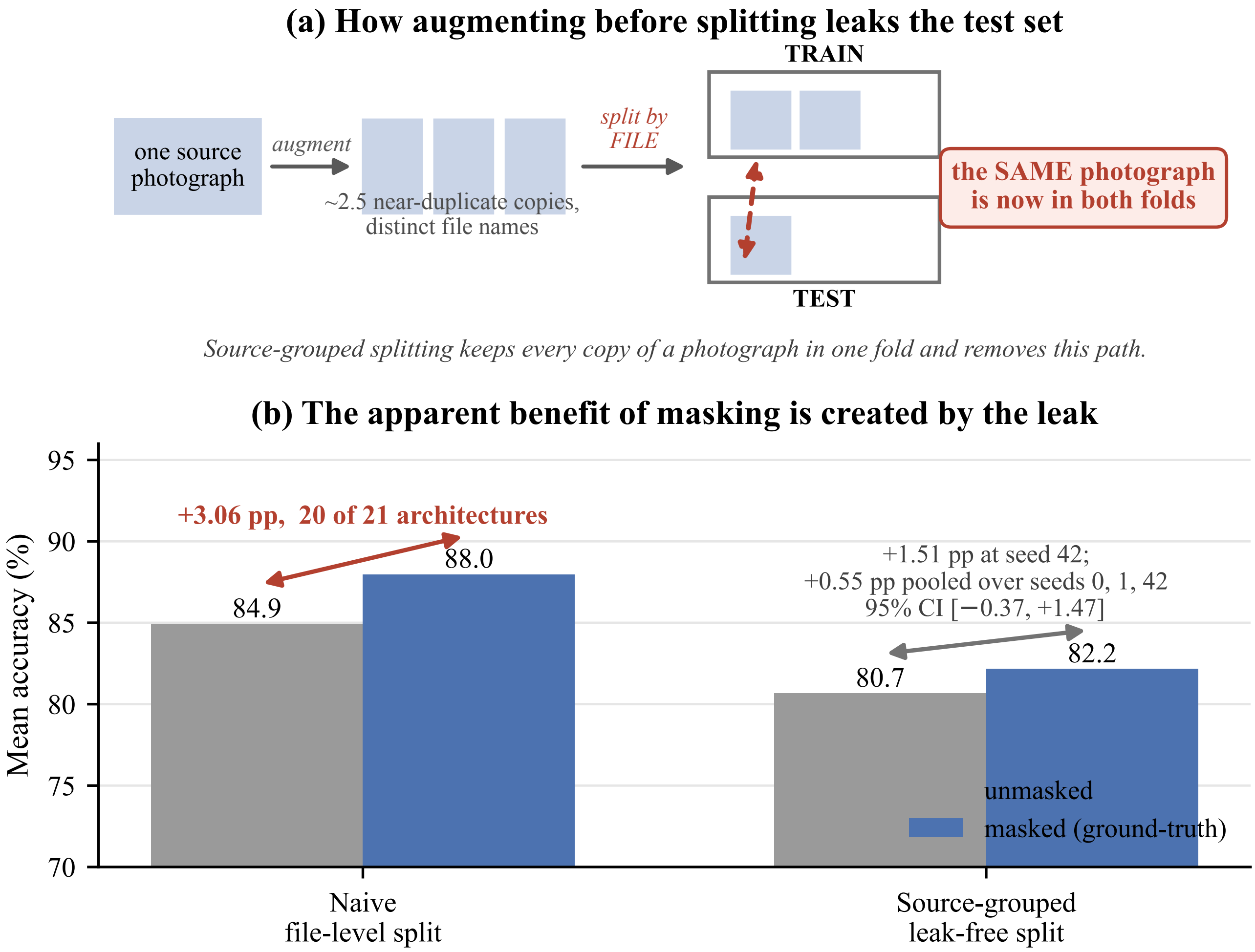


Figure 2: The data-leakage artifact at the centre of this study. (a) The source collection augments each photograph into about 2.5 near-duplicate copies with distinct file names, so partitioning the augmented files at random places copies of one photograph in both the training and the test fold. (b) The consequence, averaged over architectures: under the naive file-level split the masked condition appears to beat the unmasked one by 3.06 percentage points in 20 of 21 architectures, an artifact of that shared-photograph path. Under the source-grouped split the same comparison gives 1.51 percentage points at seed 42 and only 0.55 percentage points pooled over seeds 0, 1 and 42, with a 95 percent confidence interval spanning zero. Panel (b) plots the 21-architecture figure because that is what the original experiment produced; restricted to the eight architectures common to both analyses the naive split gives 2.07 percentage points, so the matched before-and-after contrast is $2.07 \rightarrow 0.55$. Either way, the effect that motivated the original design does not survive a leak-free split.

the two is itself one of our results. Across seeds 0, 1, and 2, the standalone classifier (ConvNeXt-Large) reached **82.6** $\pm$ **0.7** percent internally and **76.6** $\pm$ **1.6** percent on the clean external set (balanced 80.1 $\pm$ 1.8). The robust pipeline reached **78.9** $\pm$ **1.0** percent internally and **73.4** $\pm$ **2.5** percent externally (balanced 77.9 $\pm$ 3.4). The classifier won on both sets in all three seeds. The all-in-one YOLOv8-seg model, evaluated at the fair confidence threshold of 0.05 but trained with a single seed, reached 78.0 percent internally (balanced 77.7) and 66.8 percent

Table 3: Masking ablation (Benchmark 1) on the leak-free split. Each condition is compared to unmasked full images on the same architectures. The masked condition uses ground-truth polygon masks, so it is an oracle upper bound. Changes are means in percentage points; the architecture count for each condition is given explicitly.

Condition	Architectures	Seed 42	Seeds 0, 1, 42	Significance
Masked (background zeroed)	11	$+1.5$	$+0.55$	Wilcoxon $p = 0.32$; CI $[-0.37, +1.47]$
Cropped to bounding box	8	$\approx +0.0$	not run	not significant
Masked and cropped	8	$\approx +0.0$	not run	not significant

The seed-42 sweep covered all 11 architectures under the masked and unmasked conditions, but only 8 of the 11 under the two cropping conditions (ConvNeXt-Tiny, Swin-Base, and Swin-Large were not run with cropping, for compute reasons). The multi-seed confirmation over seeds 0, 1, and 42 was run for the masked condition only, across 8 architectures and 24 architecture-seed pairs; the cropping conditions were evaluated at seed 42 only. The seed-42 masking effect was concentrated in transformers ($+3.1$, five of five) with roughly no change for convolutional networks ($+0.2$); the transformer-versus-convolutional difference was not significant (one-sided Mann-Whitney $p = 0.33$, two-sided $p = 0.66$). The pooled masking effect is $+0.55$ percentage points with a 95 percent confidence interval of -0.37 to $+1.47$; the interval, not a detectability threshold, is the quantity we rely on. For contrast, the naive file-level split had appeared to show $+3.06$ percentage points across 20 of 21 architectures, a leakage artifact. Restricted to the same eight architectures that enter the pooled leak-free estimate, the naive split gives $+2.07$ percentage points (7 of 8 positive), so the matched before-and-after contrast is $2.07 \to 0.55$ rather than $3.06 \to 0.55$.

externally (balanced 67.0). These are the figures for the model retrained on the source-grouped split (Section 4.11); we treat the point estimates cautiously because each is a single seed.

4.5 Giving the pipeline its best configuration

The comparison above tests the pipeline as it was built, and as built it is handicapped in a way we did not initially notice. Its classifier is trained on ground-truth masks, because that is what the classification dataset provides, but at inference it receives masks predicted by the localiser. The two distributions are not the same: predicted masks are ragged, sometimes miss part of the lesion, and occasionally cover the whole frame. A classifier trained only on perfect masks has never seen that. Attributing the pipeline's deficit to segmentation quality, as an earlier version of this analysis did, presumes the answer; the deficit could equally be a train-test mismatch.

We therefore ran the pipeline in three configurations, ten seeds each, against the same standalone classifier. Regime A is the published one, trained on ground-truth masks and tested on predicted. Regime B matches the two, training the classifier on predicted masks generated by the same localiser that serves it at test time. Regime C matches them and additionally dilates the mask by 12 pixels, retaining a rim of peri-lesional skin on the grounds that clinicians read depth partly from the tissue bordering a burn. Table 5 reports all three.

Before the comparison itself, one point of bookkeeping. Regime A is nominally the same configuration as the pipeline arm of Section 4.6, and both are paired against the same ten standalone-classifier runs, but they are different training runs of that configuration: regime A averages 81.27 percent internally and 76.33 externally, against 80.98 and 75.02 for the arm in Table 6. Two ten-seed estimates of one configuration differing by 0.29 and 1.31 percentage points is itself an instance of the run-to-run variation Section 4.9 measures. It is also why every comparison in this paper is reported as a paired difference against a classifier trained in the same campaign, and why we do not compare levels across tables.

Matching the mask source is worth $+2.05$ percentage points internally (95 percent interval $+0.11$ to $+3.99$, $p = 0.041$), which is most of the gap the published configuration showed. Dilation adds nothing beyond that internally ($+2.00$ over regime A, $p = 0.10$) and costs on the external set. The consequence for the head-to-head is substantial and we state it plainly. Against the published configuration the classifier leads internally by 2.34 points ($p = 0.025$). But **against the pipeline's best configuration the two systems are statistically indistinguishable**: internally by 0.29 points in the classifier's favour (95 percent interval -1.12 to $+1.71$, $p = 0.65$), externally by 1.79 points (-0.50 to $+4.07$, $p = 0.11$). Neither difference is resolved at ten seeds on either set.

Two readings are available and we prefer the conservative one. The generous reading is that segmentation-guided classification matches a plain classifier once it is tested fairly, and that the earlier deficit was largely an artifact of the mismatch. The conservative reading is that a tie is not a win: the pipeline attains parity while running two models instead of one, at 1.6 times the latency (Section 4.13), and parity does not justify that cost on accuracy grounds. Both readings agree on the fact, which is that the published configuration understated the design by about two points and that no configuration we tried beats the simpler alternative.

We also note what this does to the three-seed pilot that motivated the experiment. On three seeds, regime C reached 83.90 percent against the classifier's 83.61 and appeared to win. At ten seeds it is 83.27 with a standard

Table 4: Head-to-head accuracy (percent). Multi-seed rows are mean ± sample standard deviation (denominator $n-1$) over seeds 0, 1, and 2, with the Wilson 95 percent interval for the seed-42 run given beneath each. Balanced accuracy is given for the imbalanced external set. Reference rows below the rule isolate the oracle-mask upper bound and the threshold-sensitivity ablation and are not the deployed system.

System	Internal ($N=205$)	External ($N=319$)
Standalone classifier (ConvNeXt-Large)	82.6 ± 0.7	76.6 ± 1.6 (bal. 80.1 ± 1.8)
Wilson 95% CI (seed 42)	[76.6, 87.0]	[69.2, 78.8]
Source-clustered 95% CI (seed 42)	n/a	**[68.3, 80.1]**
Robust pipeline (localiser to Swin-Tiny)	78.9 ± 1.0	73.4 ± 2.5 (bal. 77.9 ± 3.4)
Wilson 95% CI (seed 42)	[75.6, 86.2]	[64.7, 74.7]
Source-clustered 95% CI (seed 42)	n/a	**[63.7, 75.7]**
Standalone YOLOv8-seg (single seed, conf. 0.05)	**78.0** (bal. 77.7)	**66.8** (bal. 67.0)
same model before the split was corrected	*90.7* (bal. *90.6*)	*48.9* (bal. *50.3*)
Oracle Swin-Tiny (ground-truth masks)	83.9	n/a
Strict pipeline (conf. 0.25, no-detection = error)	79.0	54.2
Robust pipeline (seed 42)	81.5	69.9

The standalone YOLOv8-seg row is the model retrained on the source-grouped split (Section 4.11); the italicised row beneath it is the same architecture and recipe trained on the collection's original split, which had seen 87.3 percent of the internal test set. Both are given because the pair is informative: the contaminated model's internal-to-external drop is 41.8 points against the retrained model's 11.2, and that excess is the leakage signature. The internal gap between the two rows is not attributable to leakage alone, since the corrected partition also leaves 21 percent less training data (Section 4.11). The robust and strict pipeline rows carry the *original* localiser. Re-scoring the ten-seed pipeline arm on the retrained one changes its internal accuracy by 0.10 pp (Section 4.11), so these rows are not materially inflated by that contamination; the selection effect of Section 4.12 is the live concern for this table. The oracle-mask row and the classifier rows never involve a trained segmenter and are unaffected throughout.

Two questions, two tests. *Does the classifier beat the pipeline on these particular images?* Paired exact McNemar on the seed-42 models: internal 20 versus 18 discordant pairs, $p=0.87$ (a tie; seed 42 was the pipeline's most favourable run); external 54 versus 40, $p=0.18$. *Does the classifier beat the pipeline on average across training runs?* The paired per-seed difference is $+3.74$ pp internally (95% CI $+3.04$ to $+4.44$, paired t $p=0.002$) and $+3.24$ pp externally (95% CI -3.83 to $+10.31$, $p=0.19$), with the classifier ahead in three of three seeds on both sets. The internal advantage is therefore statistically supported at the level of training runs, while the single-run image-level comparison is not resolved at $N=205$. We note the paired tests use only three seeds and are correspondingly imprecise; Table 6 repeats them over ten.

Table 5: The two-stage pipeline in three configurations, ten seeds each, against the same standalone classifier. Regime A is the published configuration, whose classifier is trained on ground-truth masks and tested on masks predicted by the localiser. Regimes B and C match the training mask source to the test mask source; C additionally dilates the mask by 12 pixels to retain peri-lesional skin. Differences are paired by seed against the classifier.

Configuration	Internal ($N=205$)	Paired difference	External ($N=319$)	Paired difference
Standalone classifier (ConvNeXt-Large)	83.61 ± 1.74	—	77.81 ± 2.29	—
A: GT-trained, predicted-tested	81.27 ± 1.74	-2.34 $p=0.025$	76.33 ± 1.23	-1.47 $p=0.085$
B: mask source matched	$\mathbf{83.32 \pm 1.50}$	$\mathbf{-0.29}$ $\mathbf{p=0.65}$	76.02 ± 1.82	-1.79 $p=0.11$
C: matched and dilated	83.27 ± 2.41	-0.34 $p=0.67$	75.24 ± 2.45	-2.57 $p=0.008$

Matching the mask source is worth $+2.05$ percentage points internally over the published configuration (95 percent interval $+0.11$ to $+3.99$, $p=0.041$); dilation adds nothing beyond that and costs externally. Under regime B neither the internal nor the external difference from the standalone classifier is resolved at ten seeds, so the two systems are statistically indistinguishable on both sets. A three-seed pilot had put regime C at 83.90 percent, apparently ahead of the classifier; ten seeds place it at 83.27 and behind.

deviation of 2.41, and it loses. That is the eighth time in this study a three-seed result did not survive ten, and on this occasion the result it offered was the one we wanted.

4.6 How many training runs the comparison needs

The two questions of Section 3.6 give different answers, and separating them matters. At the level of *these particular images*, the advantage is not resolved. On the seed-42 models McNemar's exact test gave 20 versus 18 discordant pairs internally ($p=0.87$, effectively a tie, because seed 42 happened to be the pipeline's most favourable run) and 54 versus 40 externally ($p=0.18$).

At the level of *training runs* the original three-seed protocol suggested a clear internal advantage and an unresolved external one. Neither conclusion survived more seeds in the form it was stated, and the way each failed is instructive enough to report in full. We repeated the entire head-to-head over **ten** seeds, training both arms under a fixed recipe and evaluating each on both test sets. The

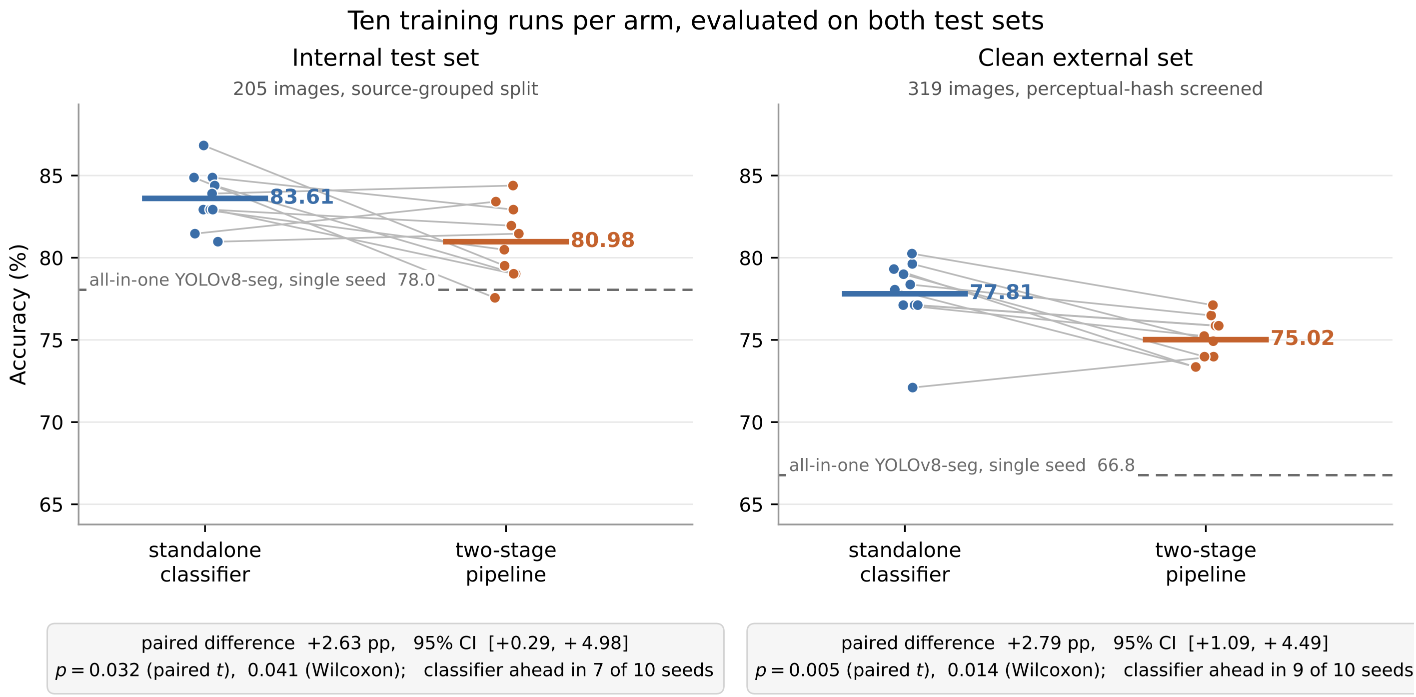


Figure 3: The head-to-head comparison, plotted one training run at a time. Each dot is a single seed; the grey line joins the same seed across the two arms, because the two arms are trained in matched pairs and the comparison is paired. The heavy bar is the mean of the ten runs. The dashed line is the all-in-one YOLOv8-seg model retrained on the source-grouped split, which is single-seed. Two things are meant to be visible. The classifier's mean is higher on both test sets, and the gap survives the paired test. But the runs overlap heavily, and internally three of the ten seeds cross: the pipeline beats the classifier in those runs. That overlap is the reason this figure plots ten seeds rather than the three of our original protocol, and it is what Section 4.9 quantifies. An earlier version of this figure showed three-seed means with standard-deviation bars, which reported an internal margin of $+3.74$ points with an interval about three times narrower than these data support.

pipeline arm used *predicted* masks from the localiser rather than ground-truth masks, so that the arm being measured is the deployed system (Table 6).

Over ten seeds the classifier leads the pipeline by **2.63** percentage points internally (95 percent CI $+0.29$ to $+4.98$; paired t $p = 0.032$, Wilcoxon $p = 0.041$; ahead in 7 of 10 runs). Externally it leads by **2.79** percentage points (95 percent CI $+1.09$ to $+4.49$; $p = 0.005$, Wilcoxon $p = 0.014$; ahead in **9 of 10** runs).

The choice of metric qualifies the internal result, and we report the qualification rather than the more convenient figure. We repeated both comparisons on *balanced* accuracy, which is the appropriate summary when classes are unequal. The external margin survives at $+1.96$ points (95 percent CI $+0.26$ to $+3.65$, $p = 0.028$, ahead in 8 of 10), but the internal margin does not: $+1.45$ points, 95 percent CI -0.74 to $+3.65$, $p = 0.17$, ahead in only 6 of 10 runs. The internal advantage is therefore a property of plain accuracy on this class distribution and is not established once the classes are weighted equally, whereas the external advantage holds under both summaries. That a metric choice of this kind can decide a conclusion is the subject of a substantial recent literature on validation practice in biomedical imaging (Maier-Hein et al., 2024), and our case is a small instance of it. This is the sixth of the evaluation choices audited in this paper (Table 12), and it points the same way as the others: the external comparison is the one that carries the argument.

The contrast with three seeds is the point. Three seeds put the internal difference at $+3.74$ points with an interval of $+3.04$ to $+4.44$ and $p = 0.002$: an interval about three times narrower than ten seeds support (1.40 against 4.69 percentage points wide), around an effect whose true magnitude is smaller. The same three seeds put the external difference at $+3.24$ points with an interval of -3.83 to $+10.31$ and $p = 0.19$, wide enough to license no claim at all, when the effect is in fact the most reliable one in the study. A three-seed protocol thus overstated the weaker result and concealed the stronger one simultaneously.

We therefore state the head-to-head as follows. *The*

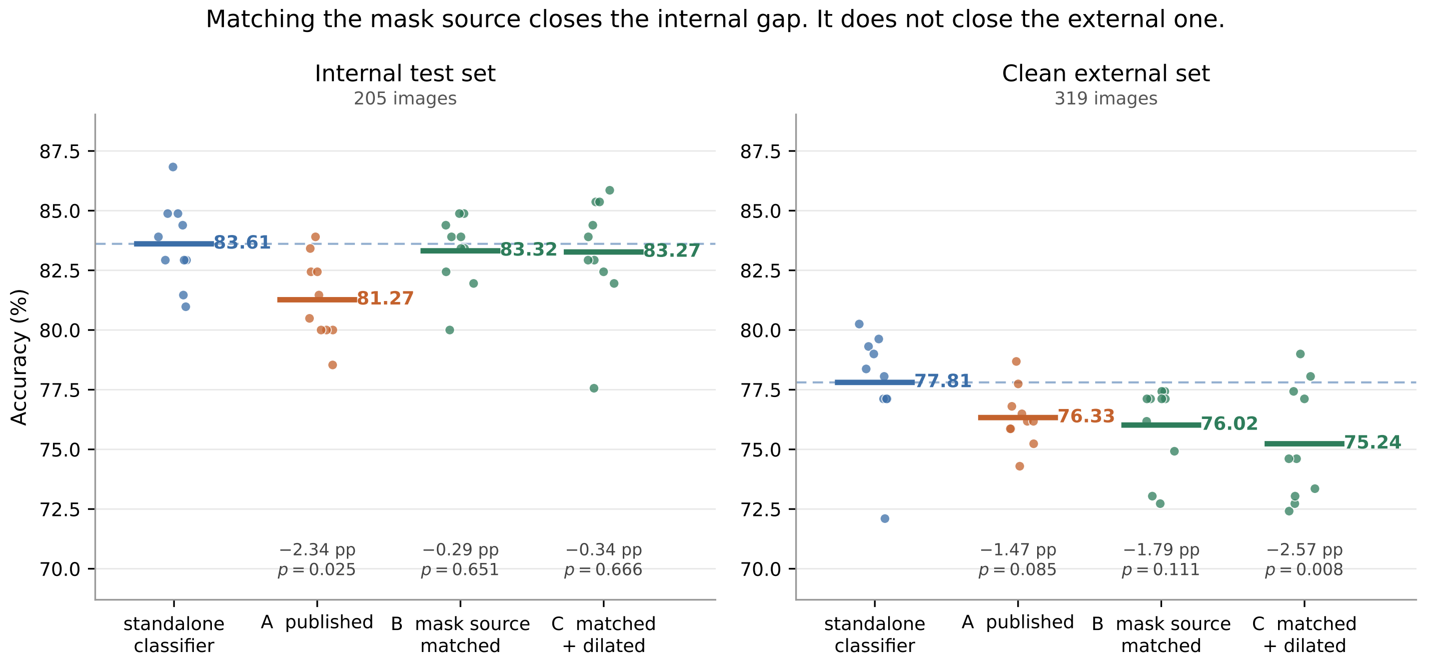


Figure 4: The pipeline as published, against the pipeline given its best configuration. Each dot is one of ten training runs; the heavy bar is the mean; the dashed line is the standalone classifier's mean. Regime A is the deployed system: its classifier is trained on ground-truth masks and then, at test time, shown masks predicted by the localiser. Regimes B and C remove that mismatch by training on predicted masks, C additionally keeping a 12-pixel rim of skin around the lesion. Internally the repair is worth most of the gap and the difference from the classifier stops being resolvable. Externally it is worth nothing: the gap widens rather than closes, and dilation makes it worse. Differences beneath each column are paired by seed against the classifier. This is Table 5 drawn.

standalone classifier outperforms the two-stage pipeline on both test sets on plain accuracy, by roughly two to three percentage points. On balanced accuracy only the external margin holds, and it is the external margin that is most reliable where it holds in nine of ten training runs. This is a more modest internal claim and a considerably stronger external one than the original protocol implied, and the external claim carries the argument, since it concerns data the model was never fitted to.

This is the clearest evidence in the paper for its own thesis. A three-seed protocol, entirely ordinary in this literature, did not merely add noise: it produced a confidently narrow interval around an overstated effect while hiding a real one. Both errors survive peer review easily, because an interval computed from three runs is typographically indistinguishable from one computed from thirty.

4.7 Oracle against real masks, and the confidence threshold

Table 4 also separates two effects that are easy to conflate. Swin-Tiny on ground-truth (oracle) masks reached 83.9 percent internally, an upper bound; the robust pipeline with real predicted masks reached 81.5 percent at the same seed and 78.9 $\pm$ 1.0 percent across seeds, so at matched seeds imperfect segmentation costs about two points. The choice of confidence threshold matters at the tail: the strict pipeline (confidence 0.25, no-detection counted as an error) reached 79.0 percent internally but only 54.2 percent externally, whereas the robust pipeline reached 69.9 percent externally at seed 42. We report both, and we never present the 83.9 percent oracle number as the accuracy of the deployed pipeline.

4.8 External validation

On genuinely clinical images the model does not transfer, and it fails in the dangerous direction. On the 94 BIP_US images (Table 8), the deployed pipeline graded about 55 of 94 images below their true severity and classified about half of the full-thickness burns as first degree. Balanced accuracy was 30.5 percent under the deep-dermal-to-second-degree mapping and 35.0 percent under the deep-dermal-to-third-degree mapping, against a chance level of 50 percent for this two-class probe.

Below-chance performance invites a mechanical explanation and we tested one rather than leaving it to a reader. The BIP_US reference contains no first-degree category,

Table 6: The head-to-head repeated over ten seeds, both arms trained under a fixed recipe and each evaluated on both test sets. The pipeline arm uses predicted masks from the localiser (confidence 0.05 with full-frame fallback), so it is the deployed system rather than the oracle-mask upper bound. Differences are paired by seed. The three-seed rows are the original protocol, shown for comparison.

	Classifier	Pipeline	Paired difference (95% CI)	p
Internal test ($N = 205$)				
Three seeds (original)	82.6 ± 0.7	78.9 ± 1.0	$+3.74$ $[+3.04, +4.44]$	0.002
Ten seeds	83.6 ± 1.7	81.0 ± 2.2	$\mathbf{+2.63}$ $[\mathbf{+0.29}, \mathbf{+4.98}]$	0.032
External test ($N = 319$)				
Three seeds (original)	76.6 ± 1.6	73.4 ± 2.5	$+3.24$ $[-3.83, +10.31]$	0.19
Ten seeds	77.8 ± 2.3	75.0 ± 1.3	$\mathbf{+2.79}$ $[\mathbf{+1.09}, \mathbf{+4.49}]$	**0.005**

Over ten seeds the classifier leads externally in 9 of 10 runs (Wilcoxon $p = 0.014$) and internally in 7 of 10 ($p = 0.041$). Relative to three seeds, the internal interval widens by a factor of about three and the external interval narrows by a factor of about four. Seeds 0–6 used batch 16 at learning rate 10^{-4} and seeds 7–9 batch 32 at 2×10^{-4}, a change made for compute reasons. The paired differences from the two groups are similar (internal $+2.44$ against $+3.09$; external $+2.87$ against $+2.61$; Welch $p = 0.84$ and $p = 0.92$), and we pool them on that basis. We state this as an assumption rather than as something a test established: with seven runs against three the Welch test has almost no power, and by the standard this paper applies elsewhere a non-significant result is not evidence of equivalence. Table 7 prints every run so that a reader can inspect the two groups directly. Absolute accuracies differ slightly from Table 4 because this replication uses one recipe for both arms rather than the per-arm settings of the original run; the paired differences, not the levels, are the quantity of interest.

Table 7: Every run behind Table 6. Accuracies are percentages; the difference columns are classifier minus pipeline, paired within a seed. Seeds 0–6 and 7–9 used different batch sizes and learning rates, so the two groups can be inspected separately.

	Internal			External		
Seed	Clf	Pipe	Diff	Clf	Pipe	Diff
0	84.9	82.9	$+1.95$	79.6	74.9	$+4.70$
1	82.9	80.5	$+2.44$	77.1	75.2	$+1.88$
2	84.4	79.0	$+5.37$	77.1	75.9	$+1.25$
3	82.9	82.0	$+0.98$	78.4	76.5	$+1.88$
4	84.9	77.6	$+7.32$	79.3	73.4	$+5.96$
5	81.0	81.5	-0.49	77.1	75.9	$+1.25$
6	83.9	84.4	-0.49	80.3	77.1	$+3.13$
7	82.9	79.0	$+3.90$	72.1	74.0	-1.88
8	81.5	83.4	-1.95	78.1	73.4	$+4.70$
9	86.8	79.5	$+7.32$	79.0	74.0	$+5.02$
Mean	83.6	81.0	$+2.63$	77.8	75.0	$+2.79$

yet the pipeline assigns first degree to **48 of the 94 images**, 51 percent, a class that cannot be correct under either mapping. That alone drives the balanced accuracy beneath chance. Restricting the comparison to the 46 images on which the model does commit to a class the reference can contain, balanced accuracy rises to 62.2 and 55.4 percent under the two mappings, and collapsing first-degree calls into second degree gives 56.1 and 50.5 percent. The right reading is therefore not that the model cannot rank burn depth, but that it applies a severity scale systematically offset from clinical depth staging, and offset in the dangerous direction. Assigning the mildest available grade to half of a set containing no mild burns is under-grading in its most extreme form. The transfer failure is real and the direction is the harmful one; what the below-chance figure measures is the offset rather than an inability to discriminate. On the clean external set of 319 images, the classifier was the most robust approach, with no undetected images, while the segmentation-dependent approaches degraded more (Table 4; Figure 6 shows the pipeline confusion matrices for both sets). Under-grading must be quantified conditionally, because a first-degree burn cannot be under-graded and a third-degree burn cannot be over-graded, so raw error counts are confounded by class balance. Among images that *could* be under-graded, the pipeline under-graded 19.3 percent internally (23 of 119) and **28.3 percent externally (51 of 180)**; among images that could be over-graded it over-graded 10.1 percent internally (15 of 149) and 15.2 percent externally (45 of 296). The bias toward lower severity is therefore present on both sets and is not an artifact of the external class distribution. We had described the external rate as markedly higher; testing it rather than asserting it, the two-proportion comparison gives $z = 1.77$, $p = 0.08$, and that figure ignores the source clustering of Section 3.6 and is therefore optimistic. The correct statement is that under-grading is substantial on both sets, numerically higher externally, and not established as different between them at this sample size. The perceptual-hash check that produced this set had already removed 118 training-identical images and 28 percent of sources before evaluation, so these numbers are measured on genuinely unseen data. Figure 5 shows what these errors look like in practice. The message is that the model does not generalise to clinical data, and that under-grading, directing a severe burn toward conservative care, is the more dangerous error.

Table 8: External validation of the deployed two-stage pipeline on the BIP_US clinical database (94 images). Values are the number of images the pipeline predicted in each class, grouped by ground-truth burn depth. About 55 of 94 images were under-graded and about half of the full-thickness burns were called first degree.

Ground truth (depth)	Predicted first	Predicted second	Predicted third	n
Superficial dermal	13	27	2	42
Deep dermal	25	7	0	32
Full thickness	10	7	3	20

Balanced accuracy 30.5 percent under the deep-dermal-to-second-degree mapping and 35.0 percent under the deep-dermal-to-third-degree mapping, against a chance level of 50 percent for this two-class probe.

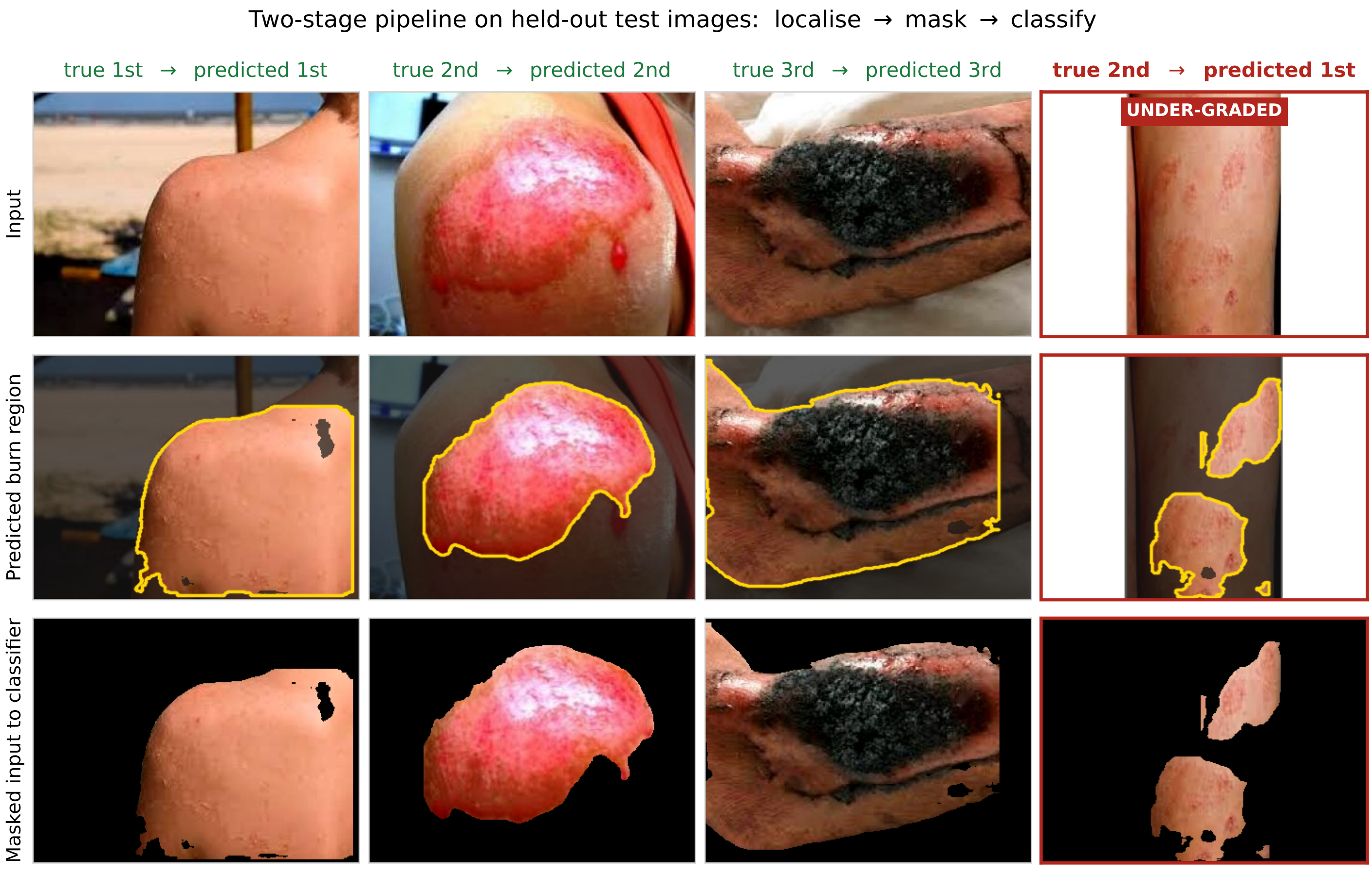
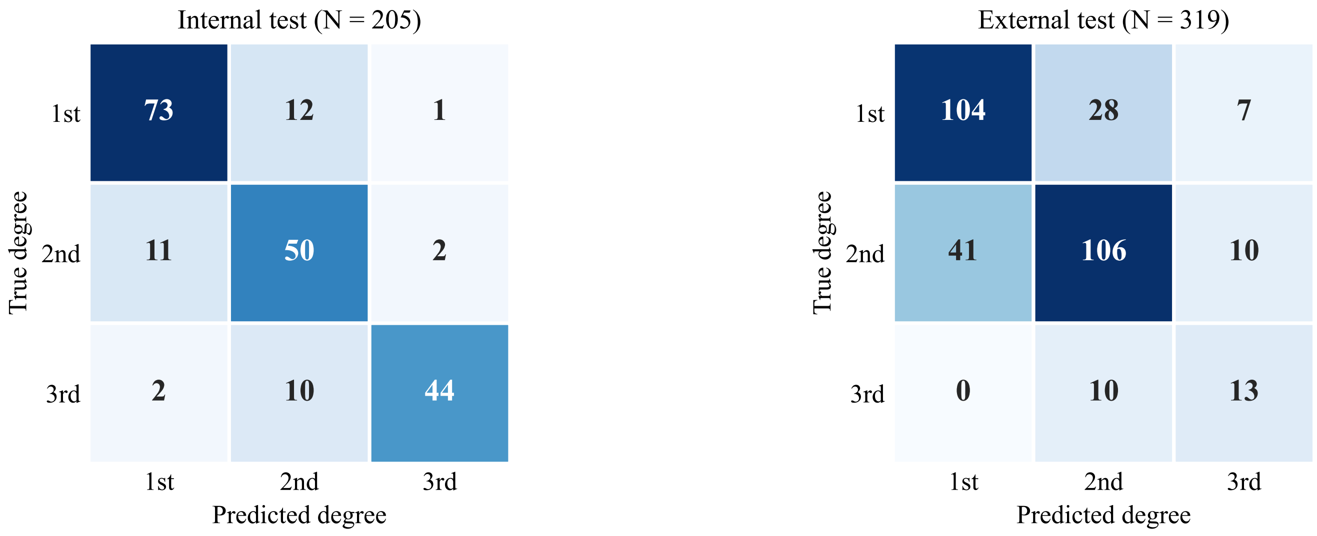


Figure 5: Qualitative behaviour of the deployed two-stage pipeline on held-out internal test images, one per column: the input photograph, the region the localiser predicts (outlined, with everything outside it dimmed), and the masked image the classifier actually receives. The first three columns are correctly graded examples spanning the three severity classes. Note how much of each photograph the mask discards. The fourth column is a representative failure of the kind that matters clinically. The burn is second degree, but its lesions are sparse and disconnected; the localiser recovers only two of them, so the classifier is handed a few fragments of skin and none of the surrounding limb, and it grades the image first degree. The plain classifier, given the whole photograph, grades the same image correctly. Errors in this direction, toward lower severity, would route a patient toward conservative care, and they are the failure mode quantified in Section 4.8.

4.9 The range of answers any three of these ten seeds would give

Every multi-seed figure in the original protocol rests on three training runs, which is common practice but a thin basis for a variance estimate. The ten-seed replication of Section 4.6 lets us measure directly how thin (Table 9).

Over ten seeds the standalone classifier's internal accuracy has a sample standard deviation of **1.74** percentage points and a range of 5.85 points; the pipeline's is 2.22 points with a range of 6.83. Both are far larger than the three-seed protocol reported (0.75 and 1.02). The reason is visible directly: across all 120 possible three-seed subsets of these ten runs, the estimated standard deviation for the



(a) (b)

Figure 6: Confusion matrices for the robust two-stage pipeline on a representative seed. Internally (a) errors fall mainly between neighbouring degrees, with second degree the hardest class, consistent with the segmentation result. On the clean external set (b) accuracy is lower and the more severe burns are frequently under-graded, for example second degree predicted as first, the dangerous direction discussed in Section 4.8.

classifier ranges from **0.00 to 3.25** percentage points, and for the pipeline from 0.28 to 3.69. A three-seed standard deviation is therefore not a noisy estimate of the right quantity; it is very nearly uninformative, and it can be arbitrarily small purely by chance. Any interval built on one inherits that defect, which is exactly what Section 4.6 observed.

Section 4.6 compared three seeds against ten, but those two protocols also differ in training recipe, so that comparison confounds seed count with the recipe change. The confound is avoidable, because the ten runs already contain $\binom{10}{3} = 120$ distinct three-seed subsets, and we can ask what each of them would have reported about exactly the same data. On the external comparison, ten seeds give $+2.79$ percentage points with a 95 percent interval of $+1.09$ to $+4.49$ and $p = 0.005$. The 120 subsets give point estimates from $+0.21$ to $+5.22$, p-values from 0.0005 to 0.86, and interval half-widths from 0.45 to 10.63 percentage points, a more than twenty-fold spread in apparent precision drawn from one set of runs. Only **15 of the 120** reach $p < 0.05$; the remaining 105 would have reported no external difference. Internally the spread is wider still: point estimates run from -0.98 to $+6.67$ percentage points, so some three-seed subsets place the *pipeline* ahead of the classifier, and only 6 of 120 reach $p < 0.05$. Because every one of these subsets is drawn from the same ten runs, the spread cannot come from a difference of protocol between them: it is variation in seed selection alone, which is what Section 4.6's comparison could not isolate. One qualification is owed. These ten runs are not a single recipe, since seeds 0–6 and seeds 7–9 differ in batch size and learning rate (Table 6), so a three-seed subset also varies in how it samples those two groups. The group-wise paired differences are similar and we pool them on that basis, but we state it rather than claim the analysis is confound-free.

One observation cuts against us and we report it. Our original three-seed protocol reported an internal p of 0.002, smaller than any of the 120 subsets produces (the minimum is 0.0094). The original run and the replication differ in training recipe as well as in the runs themselves, so this is not a like-for-like comparison and we draw no inference from the gap. What it does establish is that an internal p of 0.002 is not a stable property of this comparison, and we do not treat it as one.

We report this against our own results deliberately. It is the third of the seven instances of the pattern this paper is about: a routine methodological shortcut, here too few seeds, manufactures apparent precision in the same way that splitting after augmentation manufactured an apparent effect. Section 4.6 shows what it costs in practice rather than in principle, and the practical recommendation is concrete: report the number of runs, prefer intervals on paired differences to error bars on individual systems, and budget for substantially more seeds than three.

4.10 Skin tone

The probe of Section 3.7 returned no evidence that either system's accuracy depends on apparent skin tone in a way that holds across datasets (Table 11). Accuracy does not order monotonically with ITA on either test set, and the

Table 9: Seed variance measured over ten training runs on the internal test set, and how well three runs estimate it. The final column reports, across all 120 three-seed subsets of the ten runs, the smallest and largest standard deviation such a subset would have produced.

System	Seeds	Mean	Sample SD	SD from any 3 of these seeds
Standalone classifier (ConvNeXt-Large)	10	83.61	**1.74**	0.00 to 3.25
Two-stage pipeline (localiser to Swin-Tiny)	10	80.98	**2.22**	0.28 to 3.69

For comparison, the three-seed standard deviations reported in Table 4 are 0.75 and 1.02 percentage points, roughly *half* the ten-seed values of 1.74 and 2.22. A three-seed estimate can be arbitrarily small by chance: for the classifier, some three-seed subsets of these ten runs give a standard deviation of zero.

Table 10: Inference efficiency on an NVIDIA A100 (640×640 input, batch size 1). The full pipeline is about 1.6 times slower than the standalone classifier because it runs two models.

Stage or model	ms / image	Images/s
YOLO localiser (single class)	15.4	65
Standalone YOLOv8-seg (three class)	15.1	66
Swin-Tiny classifier	23.7	42
ConvNeXt-Large classifier	25.8	39
Full pipeline (localiser, mask, Swin)	42.3	24

band that performs best internally, very light at 97.6 percent, is among the weakest externally at 65.1 percent, close to the external minimum of 63.8 percent in the intermediate band. That is the behaviour of noise rather than of a skin-tone effect.

We nonetheless report in full a subgroup result that we could easily have published as a positive finding, because how it failed is the point. Restricting the internal set to first-degree burns, the standalone classifier's errors fell on markedly darker-appearing skin than its correct predictions. Median ITA was **19.2** for the 21 errors against **38.4** for the 60 correct predictions, a 19-point gap that crosses a band boundary, from tan into intermediate, with Mann-Whitney $p = \mathbf{0.0025}$ and a rank-biserial correlation of -0.45. That is a large effect in the direction the fairness literature would predict and worry about, and it survives Bonferroni correction across all 15 tests in this probe (threshold $0.05/15 = 0.0033$). Reported alone, it would read as clear evidence that the classifier is biased against darker skin.

It does not replicate. The identical test on the external set, on the same arm and the same class, gives $p = \mathbf{0.35}$ with the effect in the *opposite* direction (rank-biserial $+0.10$; median ITA 39.2 for errors against 37.6 for correct predictions). This is not a power failure: the external cell contains **139** images against the internal cell's 81, so the replication attempt was the better-powered of the two. No other cell in the probe approaches significance.

The internal result is also not stable across training runs, and the way it is unstable is worth separating from the way it is not. Repeating the same first-degree test on each of the ten classifier seeds of Section 4.6, only **3 of 10** reach $p < 0.05$; the p-values run from 0.025 to 0.72 with a median of 0.20, so the seed we happened to report was among the more extreme. Whether this subgroup effect is "significant" is therefore a property of the seed, exactly as the masking effect was. But the *direction* is another matter: in **10 of 10** seeds the errors fall on darker-appearing skin than the correct predictions. We do not convert that into a sign test, because ten classifiers trained on the same images and scored on the same 205 photographs produce heavily correlated errors and the runs are not independent draws. We report the consistency because suppressing it would be the same selective reporting we criticise, and we decline to claim it, because a directional tendency that no single run resolves and that reverses on a better-powered external set is not evidence of a fairness problem. It is a reason to look properly, with recorded phototype and a pre-registered analysis.

We draw the conservative conclusion, which is that this study provides *no* usable evidence about skin-tone fairness in either direction, and that the internal subgroup result should be read as a false positive. We report it rather than delete it for the same reason we report the leakage artifact. This is the fourth time in this paper that a finding significant under one protocol has evaporated under a second look. It is the most seductive of the seven, because a bias result is one that a reader wants to believe and that a reviewer is unlikely to challenge. A single-dataset fairness analysis is subject to exactly the same replication failure as a single-dataset accuracy claim, and the fairness setting supplies extra incentive not to look twice. Establishing whether these models are equitable requires data with recorded phototype and a pre-registered analysis, neither of which we have.

4.11 Propagating the split correction to the segmentation stage

The four audits above were deliberate. This one was not: we found it while preparing the released code, after every other number in this paper was final, and it is the most uncomfortable result here because it is the paper's own subject matter applied to the paper's own pipeline.

Table 11: Skin-tone probe. Standalone classifier accuracy by Individual Typology Angle band, with Wilson 95 percent intervals, on both test sets. Bands are ordered lightest to darkest. Accuracy does not order monotonically with apparent skin tone on either set, and the band that is strongest internally is the weakest externally.

ITA band	Internal ($n = 194$)		External ($n = 319$)	
	n	Accuracy (95% CI)	n	Accuracy (95% CI)
Very light (> 55)	41	97.6 [87.4, 99.6]	63	65.1 [52.8, 75.7]
Light (41 to 55)	24	87.5 [69.0, 95.7]	69	84.1 [73.7, 90.9]
Intermediate (28 to 41)	40	72.5 [57.2, 83.9]	58	63.8 [50.9, 74.9]
Tan (10 to 28)	34	79.4 [63.2, 89.7]	71	77.5 [66.5, 85.6]
Brown (-30 to 10)	46	78.3 [64.4, 87.7]	53	79.2 [66.5, 88.0]
Dark (< -30)	9	77.8 [45.3, 93.7]	5	80.0 [37.6, 96.4]

Tested on the continuous ITA rather than on these bands, the classifier's errors fell on darker-appearing skin internally (median ITA 22.5 for errors against 34.2 for correct predictions, Mann-Whitney $p = 0.015$) but not externally ($p = 0.35$). The two-stage pipeline showed no association on either set ($p = 0.42$ and $p = 0.76$). Section 4.10 reports the subgroup analysis and its failure to replicate. The darkest band contains 9 and 5 images respectively; no claim about it is supportable, and we report it only so that the sparsity is visible.

Correcting the split (Section 3.2) fixed the classification datasets. It did not touch the segmentation models, which had been trained earlier on the collection's original split, and we did not think to check whether the two splits agreed. They do not: **179 of the 205 internal classification test images, 87.3 percent, lie in the segmentation models' training or validation folds**. Every internal number in this paper that passes through a YOLO model is therefore measured partly on images that model was fitted to.

We quantified the consequence rather than estimating it. Re-evaluating the contaminated three-class standalone model at the fair threshold of 0.05 gives **90.7 percent** on the internal test set (balanced 90.6) against **48.9 percent** externally (balanced 50.3). A 41.8-point internal-to-external collapse is not a distribution shift of that magnitude; it is the signature of a model being scored on its own training data.

We then measured the leak directly by removing it. We rebuilt both segmentation datasets on the classifiers' source-grouped split, so that the segmentation test fold *is* the 205-image classification test fold and no source photograph appears in two folds. We then retrained both YOLOv8x-seg models from the public initialisation under the original recipe: 100 epochs at 640×640, batch 8, AdamW at 10^{-3}, seed 42, patience 20. Nothing changed except the partition. On the same 205 images, the three-class standalone model falls from 90.7 percent to **78.0 percent** (balanced 77.7, one image undetected), a paired difference that is highly significant on those images (exact McNemar, 33 against 7 discordant, $p = 4 \times 10^{-5}$). The retrained single-class localiser reaches a mask mAP50 of 0.695 on that leak-free test fold and returns a region for all 205 images, so the robust pipeline's full-frame fallback never fires internally; the retrained three-class model reaches 0.566.

That 12.7-point gap is not purely the leak, and we would be repeating this paper's own error if we reported it as though it were. Correcting the split also shrank the training set, because the source-grouped partition assigns fewer images to training than the collection's original 90/6/4 split did: 2,425 against 3,081, a reduction of 21 percent. Some of the drop is therefore the loss of training data rather than the loss of an unfair advantage. The 26 test images that *neither* model was trained on give a control for exactly this: on those, the contaminated model still scores 88.5 percent against the retrained model's 80.8, a gap of 7.7 points that cannot be leakage because neither model had seen them. On the 179 images the contaminated model had seen, the gap is 13.4 points. The difference between those two gaps, about 5.7 points, is the part attributable to leakage; but the control rests on 26 images and its bootstrap interval runs from 0.0 to 19.2 points, which is too wide to separate the two mechanisms cleanly. **What we can say is that removing the contamination cost 12.7 points overall, that a training-set reduction confounds it, and that a properly powered separation would need the leak-free model retrained at the original training-set size**. The source-grouped split makes that impossible without collecting more photographs.

The standalone-YOLO row in Table 4 can therefore be stated honestly at 78.0 percent, which places it below both the plain classifier and the pipeline and leaves the paper's ordering intact.

That left the two-stage pipeline arm, whose localiser carried the same contamination, and we closed it rather than reasoning about it. The pipeline's Swin-Tiny classifier is trained on ground-truth masks and meets the localiser only at inference, so the correction amounts to a change of test-time masks. We retrained the ten Swin-Tiny classifiers under the original per-seed recipe and re-scored all ten runs using masks from the *retrained* localiser. **The effect is negligible.** The pipeline's internal accuracy moves from

80.98 ± 2.22 to 80.88 ± 1.50 percent, and the paired margin in favour of the plain classifier from $+2.63$ (95 percent interval $+0.29$ to $+4.98$, $p = 0.032$, ahead in 7 of 10 runs) to $+2.73$ ($+0.64$ to $+4.82$, $p = 0.016$, ahead in 8 of 10).

We had expected the opposite and had said so, and being wrong here is worth reporting precisely. A leak that cost the standalone segmentation model 12.7 percentage points cost the two-stage pipeline about a tenth of a point. In the pipeline the grading is done by a classifier that was never contaminated, and the localiser contributes only a mask. Contamination does not propagate uniformly through a pipeline: where it lands depends on which stage the metric actually depends on. The consequence for this paper is that the internal head-to-head no longer carries a localiser-leak caveat. On balanced accuracy the corrected margin is $+1.65$ (-0.39 to $+3.68$, $p = 0.10$), so the metric-dependence of Section 4.6 survives the correction unchanged.

Three things are *not* affected, and we checked each rather than assuming it. The segmentation quality figures of Section 4.1 are computed on the segmentation models' own held-out split, which has zero source overlap with their training fold, so the mask mAP50 values of 0.726 and 0.603 stand. The external set has zero image-level and zero source-level overlap with the segmentation training data, so every external number in this paper is clean, including the entire ten-seed head-to-head on the external set. Benchmark 1 and the oracle-mask row use ground-truth polygon masks and never invoke a trained segmenter at all, so the paper's central masking null is untouched.

What it does affect is the *internal* pipeline arm, whose localiser front-end produces better masks on these images than it would on unseen ones. That inflates the pipeline's internal accuracy. The direction matters: the pipeline is the arm that *loses*, and it loses while enjoying this advantage. The internal margin of $+2.63$ percentage points in favour of the plain classifier is therefore a *lower bound* on the true margin, and correcting the leak would widen it. The paper's conclusion survives, but it survives as a conservative statement rather than a clean one, and a reader is entitled to know the difference.

The transferable lesson is the one we most wish we had not had to learn from ourselves. A pipeline is only as leak-free as its leakiest stage, and correcting a split for one component does not correct it for a component trained earlier under a different regime. We had written a paper arguing that leakage is easy to introduce and hard to see, had built a source-grouped split specifically to eliminate it, and had audited that split with perceptual hashing. We still shipped a two-stage system whose first stage had seen 87 percent of the test set. Nothing in our own procedure would have caught it; the check that did was comparing two directory listings.

4.12 How the comparison arm was chosen

The seventh audit is the one that costs us most, and like the fifth it was not planned. Section 3.4 says the standalone classifier is “evaluated in its best Benchmark 1 condition, ConvNeXt-Large on unmasked images”. Asked which set that judgement was made on, the answer is the test set.

Ranking the eleven unmasked architectures by *validation* accuracy instead puts Swin-Tiny first at 85.44 percent, with ConvNeXt-Large fourth at 84.47. On test the order reverses: ConvNeXt-Large reaches 84.88 percent and Swin-Tiny 82.44. We selected the architecture that won on the set we then reported it on, across a pool of eleven, which is a selection effect of exactly the kind this paper spends its length warning about.

The cost is large. Against the seed-42 robust pipeline at 81.46 percent, the test-selected arm gives a margin of $+3.41$ percentage points and the validation-selected arm gives $+0.98$: **choosing honestly removes 71 percent of the internal margin**. Five of the eleven architectures beat the pipeline on test at all, and their test accuracies span 76.10 to 84.88, so the choice of arm moves the comparison by more than the comparison itself measures.

We report this rather than quietly re-running with Swin-Tiny for three reasons. It is the correct disclosure. It is the seventh instance of this paper's own pattern, and the one that most directly implicates a headline number rather than a side analysis. And its direction is opposite to that of the fifth audit, where a leaking localiser inflated the arm that loses: one bias favours the classifier and the other favours the pipeline, they do not cancel in any principled way, and we can bound neither jointly. The consequence is stated plainly in Section 5.4: the internal head-to-head should not be read as an effect estimate. The external comparison inherits the same architecture choice and is weakened by it too, though it remains the result made on data neither arm was selected against.

4.13 Efficiency

The pipeline is slower and no more accurate than the standalone classifier (Table 10). On the A100, the localiser took 15.4 ms per image and the standalone classifiers 25.8 ms (ConvNeXt-Large) and 23.7 ms (Swin-Tiny), while the full pipeline took 42.3 ms, about 1.6 times slower than the standalone classifier because it runs two models. The pipeline's return for this cost is localisation, not accuracy. These figures are measured on a datacentre A100 and are therefore not a mobile latency budget. We did not measure on-device inference time, and the shipped demonstration application runs a Swin-Small classifier rather than the benchmarked Swin-Tiny, so no number in this paper describes what a user of the application experiences.

Table 12: Sensitivity of this study's conclusions to seven evaluation choices. Each row gives the answer a conventional protocol returns and the answer the corrected protocol returns on the same data. No new data were collected between the two columns. Rows 1, 3, 5 and 6 recompute on data already in hand; row 4 corrects by comparison against an independent set, which is itself the audited failure there; row 2 is a content-based audit of the same split. Here and in the tables that follow, pp abbreviates percentage points.

Evaluation choice	**Conventional protocol**	**Corrected protocol**	**Section**
Split the augmented files at random	Masking helps: $+2.07$ pp on the 8 architectures common to both analyses ($+3.06$ across all 21 originally trained, 20 of 21 positive)	$+0.55$ pp, 95% CI $[-0.37, +1.47]$, Wilcoxon $p = 0.32$, 13 of 24 architecture-seed pairs positive; ground-truth masks, so an upper bound	4.2, 4.3
Group by source identifier and assume the folds are independent	The folds are independent	2 of 205 internal test images (1.0%) are perceptually identical to a training image filed under a different identifier, one pair with conflicting labels; on a second public collection, 118 images and 56 of 199 sources (28%) were duplicates of our training data	3.2, 3.6
Use three training seeds	Internal $+3.74$ $[+3.04, +4.44]$, $p = 0.002$; external $+3.24$ $[-3.83, +10.31]$, $p = 0.19$	Ten seeds: internal $+2.63$ $[+0.29, +4.98]$, $p = 0.032$; external $+2.79$ $[+1.09, +4.49]$, $p = 0.005$, ahead in 9 of 10. Across all 120 three-seed subsets of those ten runs: SD 0.00 to 3.25 pp, external p from 0.0005 to 0.86, only 15 of 120 reach $p < 0.05$	4.6, 4.9
Test a subgroup on one dataset	Errors fall on darker-appearing skin: median ITA 19.2 against 38.4, $p = 0.0025$, surviving Bonferroni correction	External replication gives $p = 0.35$ with the effect reversed (rank-biserial $+0.10$) on a larger cell, 139 images against 81	4.10
Summarise accuracy the usual way	The classifier beats the pipeline internally: $+2.63$ pp $[+0.29, +4.98]$, $p = 0.032$, over ten seeds	On *balanced* accuracy over the same ten runs the internal margin is $+1.45$ pp $[-0.74, +3.65]$, $p = 0.17$, ahead in 6 of 10, and is not established. The external margin survives both summaries ($+1.96$ $[+0.26, +3.65]$, $p = 0.028$)	4.6
Take the best architecture as the comparison arm	ConvNeXt-Large is the best Benchmark 1 unmasked condition, at 84.88% internally	Best *on test*. On validation Swin-Tiny wins (85.44 against 84.47) and scores 82.44 on test. Against the seed-42 pipeline the margin falls from $+3.41$ pp to $+0.98$: honest selection removes **71%** of the internal margin	4.12
Correct the split for one stage and assume the pipeline is clean	The head-to-head is leak-free, because the classifier split is grouped by source photograph	The segmentation models were trained earlier on the collection's original split: 179 of 205 internal test images (87.3%) lie in their training or validation folds. Retraining both models on the source-grouped split drops the standalone model from 90.7% to **78.0%** on the same images (exact McNemar $p = 4 \times 10^{-5}$); that 12.7 pp is confounded with a 21% smaller training set. Re-scoring the two-stage pipeline on the retrained localiser, however, moves it only from 80.98 to 80.88%: the leak did not propagate to the arm that matters	4.11

5. Discussion

The claim this paper is organised around is that **the evaluation protocol, not the architecture, determined the conclusion**, and the sections below are the evidence for it. This study set out to test whether isolating the burn improves severity classification, and the answer is layered. Masking helps when the mask is good: with ground-truth masks the best masked configuration reaches 85.37 percent against the best unmasked configuration's 84.88 at seed 42, and Swin-Tiny gains 2.93 points from masking alone. Pooled across architectures and seeds, however, that benefit is $+0.55$ percentage points with a 95 percent interval of -0.37 to $+1.47$, which is consistent with no effect and rules out anything larger than about 1.5 points. Because the ablation used ground-truth masks, this is an upper bound: it is the most any segmenter could contribute, and it is small. End to end the answer turned on a configuration choice. Trained on ground-truth masks and tested on predicted ones, as originally built, the pipeline trails the plain classifier internally by 2.34 points; matched to its own test distribution it recovers most of that and the two become statistically indistinguishable on both sets. The pipeline attains parity, not advantage.

5.1 Why segmentation did not win

A null result is only useful if it comes with a mechanism. Ours has four parts, and the data separate them.

The problem masking was meant to solve was itself an artifact. The design rests on a premise: a classifier given the whole photograph will read the room instead of the wound, so removing the background should help. On the naive split that premise looked well supported, at $+3.06$ percentage points across 20 of 21 architectures. On the source-grouped split the same comparison gives $+0.55$ points, with an interval of -0.37 to $+1.47$. The shortcut the design was built to remove was, in the main, a leaked near-duplicate rather than a background cue. Whatever genuine shortcut remains is bounded below about one and a half points.

That bound is a ceiling, not an estimate. The ablation used ground-truth masks. It therefore measures what a *perfect* segmenter would contribute, which means no improvement in segmentation could have rescued the design. This is the single most useful thing the oracle condition buys: it converts "our segmenter was not good enough" from an open excuse into a testable claim, and the test rejects it.

Masking removes information as well as noise, and the two are of similar size. A mask is a destructive operation. It deletes the unburned skin next to the lesion, the extent of the injury relative to the limb, and the scale cues that come from seeing an entire hand or forearm. Figure 5 shows the cost concretely: on a second-degree burn with sparse, disconnected lesions the localiser recovers two fragments, the classifier receives those fragments with no surrounding limb, and it grades the burn first degree. Given the same photograph whole, the plain classifier grades it correctly. We tested the obvious repair and it did not work. Regime C dilates the mask by 12 pixels precisely to hand the peri-lesional skin back, and it changes nothing internally (83.27 against 83.32 percent) while costing 0.78 points externally. So the information masking destroys is not a thin rim around the lesion. It is more likely the global context — what part of the body this is, how much of it is affected — which no dilation recovers.

The largest single deficit was an engineering mismatch, not segmentation at all. The published pipeline trained its classifier on ground-truth masks and then showed it predicted masks at test time. Predicted masks are ragged, sometimes partial, and occasionally cover the whole frame. Repairing that mismatch is worth $+2.05$ percentage points internally (95 percent interval $+0.11$ to $+3.99$, $p = 0.041$), which is more than the entire oracle-mask ceiling. It is worth stressing what this means. Most of what looked like evidence that segmentation-guided classification is worse was not about segmentation. It was about training one stage on a distribution the other stage does not produce.

Put together: the benefit available was small even in the best case, the cost of masking was of the same order, and the configuration we had published added a separate two-point handicap on top. Correcting the last of these brings the pipeline to parity. Nothing in our results suggests a route past parity, because the ceiling measured with perfect masks is already too low for one to exist.

5.2 Seven evaluation choices, seven different failure modes

A second theme runs through the paper, and a reader cannot assess the first without it. Seven evaluation choices, each conventional in this literature, each changed the answer we obtained. Table 12 states all seven side by side, giving the answer a routine protocol returns and the answer the corrected protocol returns on the same data; Section 6 draws out what we take from them.

The point to make here is that these are not seven versions of one mistake. They come from seven different families: a split procedure, a grouping heuristic, a seed count, a subgroup test, an incompletely propagated correction, a summary metric, and a model selected on the set it was scored on. No single safeguard would have caught more than one of them. That matters for how the result should be read, because it means our experience is not a story about one careless step that others can simply avoid — it is a story about how many independent places a protocol offers for the same kind of failure.

We nonetheless keep the segmentation-first design, and it is important to say why. Its justification is not accuracy, which we are careful not to claim, but localisation and interpretability. The system returns where the burn is, and a clean masked region a clinician can inspect. That suits the mobile use-case and follows an established two-stage template (Bruno et al., 2025; Al-masni et al., 2020). Any grading improvement segmentation could offer here is bounded well below the run-to-run variation of the training procedure, so what it reliably delivers is localisation, not accuracy. The like-for-like gap at the same seed, 83.9 percent with oracle masks against 81.5 percent with real predicted masks, is the cost of imperfect masks rather than a hidden pipeline accuracy.

5.3 What this generalises to

Nothing in the seven audits is about burns. Each is a property of a study design, and each transfers directly to any imaging problem with the same design.

Any collection augmented before it is split. Augment-then-split is common wherever data are scarce, which is most of medical imaging. The failure is hard to see because the leaked items are not exact duplicates; they are rotations, crops and colour jitters of one another, and no exact-match check finds them. Dermoscopy and skin-lesion collections, histopathology tiles cut from one slide, frames sampled from one endoscopy or ultrasound video, and multiple slices of one MRI volume all have this shape. The correction is the same in each case: partition by the physical thing — the photograph, the slide, the patient, the study — and never by the file.

Any two-stage design in which a region is localised and then classified. This is the standard template for skin-cancer triage, lung-nodule characterisation, retinal-lesion grading and cell-level analysis. Two of our findings apply to all of them. The first is that the ceiling should be measured with ground-truth regions before a segmenter is blamed for a disappointing result. That is the only way to distinguish "the segmenter is weak" from "the design cannot help here." The second is that the classifier must be trained on the same kind of region it will be shown at inference. Training on annotated regions and testing on predicted ones is an easy mistake, it is rarely reported, and here it cost more than the entire effect under study.

Any comparison whose arms have different exposure to a confound. This is the part we believe is least appreciated. Leakage is usually discussed as something that inflates a score, and a score inflated uniformly across conditions leaves a comparison intact. Ours was not uniform: 80.5 percent of test images leaked in one arm and none in the other, because the two arms had been partitioned separately. The leak therefore did not inflate a number, it manufactured a *conclusion* — and it did so consistently across 20 of 21 architectures, which is exactly the pattern an author reads as robustness. When two conditions are prepared by different pipelines, consistency across models is evidence about the preparation as much as about the models. The nearest published relative we found is confound-leakage (Hamdan et al., 2023), and we scope our claim against it rather than claiming priority.

Any single-dataset subgroup finding. Our skin-tone result was large, in the direction the fairness literature predicts, and survived Bonferroni correction across fifteen tests. It reversed on a better-powered external cell. Subgroup claims are not more robust than accuracy claims merely because they are about fairness, and the field's growing appetite for them makes single-dataset replication failure a live risk rather than a hypothetical one.

The high single-dataset accuracies reported in the burn literature, 95.6 percent for a transfer-learning classifier (Suha and Sanam, 2022) and up to 98.1 percent among the studies a recent review collates (Fu, 2026), are of the kind our own leaky configurations produced before correction. That is exactly why such numbers deserve caution when the dataset is augmented and the split procedure is not described. The mirror image is instructive: multi-source training improves robustness (Elsarta et al., 2025), whereas our single-source model fails to transfer. Our external failure is consistent with a wide medical-imaging pattern in which strong internal scores collapse on external data (Zech et al., 2018; Yu et al., 2022; Mårtensson et al., 2020; Roberts et al., 2021).

External behaviour is at once the strongest limitation and the strongest message, and it differs by set. On the genuinely clinical Seville images the model applies a severity scale offset from clinical depth staging, assigning the mildest grade to half of a set that contains no mild burns. On the second, web-sourced external set the decline is more modest and the plain classifier remains the most robust approach, consistent with a smaller distribution shift for web images than for clinical photographs. In neither case is the system ready for a clinical role. The clinical collapse most likely reflects a narrow, web-sourced training distribution and a labelling protocol that differs from clinical depth staging. Multi-source, clinically labelled data and domain-adaptation methods are needed before generalisation can be claimed.

5.4 Limitations

Several limitations qualify these results. **The internal head-to-head should not be read as an effect estimate.** The classifier arm was selected as the best of eleven architectures on the test set, and selecting on validation instead removes 71 percent of the seed-42 margin (Section 4.12). The margin also fails to survive a change of summary metric (Section 4.6). The localiser contamination, which we had expected to work in the opposite direction, turns out to move the pipeline by 0.10 percentage points and can be set aside (Section 4.11). What we retain from the internal comparison is that no configuration of the pipeline we tried was clearly better than a plain classifier; what we do not retain is a number. The external comparison is affected by the architecture choice but not by the localiser leak, and it is the result we rely on. Beyond that, the most serious limitation is the one in Section 4.11: our source-grouped split was built for the classifiers and never propagated to the segmentation models, so 87.3 percent of the internal test set lies in their training or validation folds. We retrained both segmentation models on the source-grouped split and then re-scored the two-stage pipeline arm on the corrected localiser, retraining its ten Swin-Tiny classifiers under the original per-seed recipe. That closes the contamination question for the head-to-head: the pipeline moves by 0.10 percentage points and the margin is, if anything, slightly better resolved (Section 4.11). The remaining threat to the internal comparison is not leakage but selection, and it is the subject of Section 4.12. The external comparison, the segmentation quality figures and the whole of Benchmark 1 are unaffected, and we verified each rather than assuming it. The training data come from a single web source with community labels that were not clinically verified. The internal test set is small ($N = 205$, one image per source), so confidence intervals are wide. The head-to-head standalone YOLOv8-seg model is single-seed, so its point estimate should not be compared to the multi-seed means without that caveat. The masking delta is not stable to the architecture pool or to run-to-run noise, so we make no claim that masking is exactly zero. The headline multi-seed figures rest on three training runs, and Section 4.9 shows directly that three runs estimate the seed variance poorly. A ten-seed replication of the classifier gave a standard deviation of 1.74 percentage points against 0.75 from three, and any three of those ten would have yielded anywhere from 0.00 to 3.25. Repeating the full head-to-head over ten seeds (Table 6) changed which comparison the evidence most strongly supports, so the three-seed intervals in Table 4 should not be read as reliable; we retain them only because they are the original protocol. Future work of this kind should budget for substantially more seeds than is conventional. The BIP_US external set is small and lacks first-degree cases, so it probes second versus third degree only. We also conducted no user study or prospective clinical evaluation of the mobile application, so its usability and clinical value remain untested. Finally, our skin-tone analysis is a proxy measurement and must not be read as a fairness audit. ITA estimated from uncontrolled photographs tracks apparent skin tone in the image rather than constitutive phototype, the darkest band holds only 9 and 5 images, and the one subgroup signal we found did not replicate (Section 4.10). Whether these models are equitable across skin tones remains genuinely unknown, and answering it needs data with recorded phototype and a pre-registered analysis.

Five further limitations are worth stating plainly, because a reader will otherwise have to infer them.

One modality, and no clinical context. Everything here is a colour photograph. We use no thermography, no hyperspectral or laser-Doppler imaging, and none of the clinical variables a burn assessment actually rests on: total body surface area, mechanism and time since injury, patient age, comorbidity, or the serial re-inspection by which depth is established in practice. A photograph-only model is therefore not attempting the clinical task; it is attempting a proxy for one part of it.

Acquisition is uncontrolled and unrecoverable. The images are web-sourced. None of the test images we sampled carries any EXIF record, and the collection has been re-encoded to a single resolution, so camera, lens, illuminant, exposure and white balance are not merely uncontrolled but unavailable for analysis after the fact. This limits the work in two specific ways. Colour is the dominant cue for burn depth and it is exactly the channel most disturbed by uncalibrated capture, and our skin-tone probe estimates ITA from those same uncalibrated pixels, which is the main reason we treat it as a proxy rather than a measurement.

Errors propagate, and a cascade cannot recover them. The two stages run in sequence and the second cannot repair the first. A region the localiser misses is not merely down-weighted, it is set to zero and the classifier never sees it. Figure 5 shows one such case, and it is a general property of hard masking rather than a quirk of this localiser. It is also the strongest argument for the jointly optimised, softly masked alternative in Section 5.5, which we did not build.

No clinician was in the loop, at any point. The severity labels are community annotations from a public collection rather than clinical depth staging, no clinician reviewed the model's outputs, and no clinician assessed the mobile application. We can therefore say what the model does relative to those labels, and nothing about whether it would help or harm a clinician using it.

One underlying collection. The training data, the internal test set and the source-grouped split all derive from a single Roboflow collection. Our two external sets probe generalisation but neither substitutes for training on multiple independent sources, and the leakage phenomenon we characterise was found in one collection's augmentation practice. We expect it to be common, since augment-then-split is common, but we have measured it once.

5.5 Future work

Our results close some directions and open others, and it is worth separating the two.

Closed by the evidence here. Improving the localiser will not rescue the design. The masking ceiling was measured with ground-truth masks and is already small, so a better segmenter has little to recover. Returning perilesional skin to the classifier will not either; regime C tested that directly and it gained nothing internally and lost externally. We report these as closed so that others do not spend compute on them.

Open, and in our view the most promising. The two stages are trained separately and never see each other's errors. Training them jointly — a segmentation loss and a classification loss optimised together, with the mask applied differentiably as a soft attention rather than a hard multiply — would let the localiser learn which regions the grader actually needs, instead of the regions a human annotator outlined. Our regime B result is a crude one-step approximation of this: it aligns the two distributions once, after the fact, and is worth two points. A jointly optimised system would align them continuously. Soft masking is attractive for the same reason, because it attenuates the background rather than deleting it, which is the failure mode Section 5.1 identifies.

Segmentation supervision is the binding constraint on data. Polygon annotation is expensive, and it is why our segmentation set is small. Semi-supervised and weakly supervised segmentation — pseudo-labelling from a small annotated core, or training from image-level severity labels alone — would let the localiser use the large pool of photographs that carry a grade but no polygon. Promptable and medical-domain foundation segmenters make this cheaper still: SAM 2 (Ravi et al., 2024) produces high-quality masks from minimal prompting, and MedSAM (Ma et al., 2024) adapts that family to medical images. Either could supply pseudo-labels at a scale hand annotation cannot reach. On the classification side, self-supervised and modernised backbones (Oquab et al., 2023; Woo et al., 2023) are the natural next comparison arms; we did not evaluate them, and we do not assume they would change the conclusion, because the ceiling we measured constrains the two-stage design rather than any particular backbone.

Segmentation uncertainty is the piece we did not measure. The localiser emits a mask and the pipeline treats it as fact. It has no way to say that it is unsure, and the classifier has no way to weight the evidence accordingly. Calibrated segmentation uncertainty would give the cascade something it currently lacks entirely: the option to abstain, or to fall back to the whole frame, on the cases where the mask is least trustworthy. Given that hard masking is where we locate the information loss, we would rank this alongside joint optimisation as the most direct test of whether the design can be rescued.

What the external results demand. None of the above addresses the finding that matters most clinically. The system does not transfer, and it fails toward lower severity. That needs multi-source, clinically labelled data with depth staging performed to a documented protocol, splits at the patient rather than the photograph level, and domain adaptation evaluated on held-out sites. It also needs prospective evaluation: no user study or clinical assessment of the mobile application has been performed, so its usability is untested. Finally, the skin-tone question remains genuinely open. Answering it requires recorded phototype rather than an image-derived proxy, adequate numbers in the darker bands, and a pre-registered analysis — three things our probe lacked, which is why we report it as a null with the signal disclosed rather than as a fairness audit.

6. What we learned

If a reader takes one thing from this paper, we would like it to be this: **leakage did not merely inflate our accuracy, it changed our conclusion.** The distinction is the whole point. A leak that inflates every arm equally leaves a comparison standing. Ours inflated one arm and not the other, because the two arms had been partitioned separately, and it produced a design recommendation — mask the image before classifying it — that a clean split does not support. The recommendation was consistent across 20 of 21 architectures, which read as robustness and was in fact the signature of a shared cause.

Six further lessons follow, and none of them is specific to burns.

Partition by the physical object, then verify it. Splitting augmented files at random is the error; grouping by source identifier is the standard fix; and the standard fix was not sufficient here, because 2 of our 205 test images were still perceptually identical to a training image filed under a different identifier, one of them with a conflicting label. Identifier grouping is a heuristic. A content-based check is what turns it into evidence.

Propagate a correction to every stage, then re-audit. We built a leak-free split for the classifiers and never applied it to the segmentation models, which left 87.3 percent of the internal test set inside their training folds. We found it while packaging the code for release, not while doing the analysis. A correction applied to part of a pipeline is a correction the rest of the pipeline will quietly ignore.

Three seeds cannot estimate a variance. Of the 120 three-seed subsets of our ten runs, only 15 detect the external effect that ten seeds resolve at $p = 0.005$, and the estimated standard deviation across those subsets ranges from 0.00 to 3.25 percentage points. A three-seed interval is not a noisy version of the right answer; it is close to uninformative, and it can be arbitrarily narrow by chance. Eight separate three-seed results in this study failed to survive ten.

Measure the ceiling before blaming the component. Running the masking ablation with ground-truth masks cost us one extra experiment and settled a question we would otherwise still be arguing about. It converts "our segmenter was not good enough" into a claim that can be rejected.

Do not select the comparison arm on the test set. Ours was chosen as the best of eleven architectures on test. Selecting on validation instead removes 71 percent of the internal margin. This is why we report no internal effect estimate at all.

A subgroup finding is not more trustworthy for being about fairness. Ours was large, correctly signed, and Bonferroni-surviving, and it reversed on a better-powered external set. We report it because deleting it would have been the selective reporting this paper argues against.

What unites these is not carelessness; each individual choice is defensible and most are conventional. What unites them is that a single evaluation cannot separate a finding from an artifact of the procedure that produced it, and that which artifacts get a second look is decided by what the investigator already believes. We checked masking and seed counts early, because we had no stake in either. We nearly did not check the skin-tone result, because it agreed with us. We did not re-audit the split, because we had already corrected it once.

7. Conclusion

We asked whether isolating the burn improves severity classification. We built a segmentation-guided pipeline to do it, and tested that pipeline against the plain classifier it is meant to improve on.

The answer is no, and the bound is tight. With perfect masks the benefit is $+0.55$ percentage points, 95 percent interval -0.37 to $+1.47$; because that ablation used ground-truth masks, it is an upper bound on what any segmenter could contribute. End to end, once the classifier is trained on the same kind of masks it is tested on, the pipeline is statistically indistinguishable from a plain classifier on both test sets. In its original mismatched configuration it trailed by 2.34 points internally, and matching the mask source recovers $+2.05$ of that. Segmentation-first is therefore justified here by localisation and interpretability, not by accuracy: parity is what it achieves, and it runs two models to achieve it. The system does not transfer to independent clinical images, where it applies a severity scale offset from clinical depth staging in the dangerous direction.

We would not have obtained that answer under the protocol we started with. Under a file-level split the same experiment says the opposite, across 20 of 21 architectures. No network changed between those two answers, which is why we state the paper's result as: **the evaluation protocol, not the architecture, determined the conclusion.** Section 6 sets out that finding and the six choices like it. What this domain needs next is multi-source, clinically labelled data with patient-level splits and domain adaptation — and to treat segmentation as a route to localisation rather than as an accuracy lever.

None of this is really about burns. Augment-then-split is standard wherever data are scarce; localise-then-classify is the standard template for skin-cancer triage, lung-nodule characterisation, retinal-lesion grading and cell-level analysis; and comparisons whose arms are prepared by different pipelines are everywhere. A leak that inflates every arm equally leaves a comparison standing, and a field that treats leakage only as score inflation will keep missing the kind we found — the kind that leaves the numbers plausible and changes the answer. We would rather this paper be read as a worked example of that failure, on a small burn dataset where we could afford to chase it down, than as a result about burn photographs.

Acknowledgments

This study began as the authors' undergraduate graduation project at the Faculty of Computer and Information Systems, Islamic University of Madinah. All three authors have since graduated; the additional work reported here, comprising the leakage audit, the leak-free re-analysis, the ten-seed replication and the external validation, was carried out independently, without institutional supervision, funding or compute. This research received no external funding.

The authors thank their supervisor Dr. Hani Sayaf Almoamari for guidance throughout the original graduation project, the Biomedical Image Processing Group of the University of Seville for making the BIP_US database available for research, and the Google Colab and Kaggle platforms for computational resources.

Declaration of generative AI and AI-assisted technologies. During the preparation of this work the authors used Anthropic Claude (Claude Code; Opus 4.8 and Opus 5 models) to organise the project materials, reproduce and check the reported results, run the statistical, benchmarking and external-validation analyses (including the leak-free re-runs, the ten-seed replication, the perceptual-hash leakage check and the skin-tone probe), and to draft and edit the manuscript text. All drafting was performed from author-supplied results and under author direction; the authors verified every quantitative claim against the underlying result files, and the repository cited in the Data Availability Statement contains a script that recomputes each published number from the raw data. No text was accepted without author review, and the tool was not used for unsupervised, de novo content creation. The research itself, comprising the system design, model training, dataset preparation and the mobile application, was conceived and carried out by the authors, who take full responsibility for the content of this publication.

Ethical Standards

This study used only publicly available, de-identified burn images together with the BIP_US research database, which was obtained from the Biomedical Image Processing Group of the University of Seville for research use. No new data were collected from human participants and no identifiable individuals appear in the reported figures. Institutional review board approval was therefore not required. The work follows appropriate ethical standards in conducting the research and in writing the manuscript. We note explicitly that the system described here is a research and demonstration artifact; the paper reports that it does not generalise to independent clinical images, and it must not be used for clinical decision-making.

Conflicts of Interest

We declare we don't have conflicts of interest. The authors have no financial, organizational, commercial or personal conflicts of interest to disclose, and have not recently collaborated with any member of the MELBA editorial board.

Data availability

All data supporting the findings of this study are available. The primary burn dataset is the publicly available "skin burn wound classification" collection on Roboflow Universe (Binus, 2023), released under a Creative Commons Attribution 4.0 licence. The BIP_US clinical database is available for research from the Biomedical Image Processing Group of the University of Seville, on request to its custodians; we redistribute no BIP_US image. Code, the leak-free split definition and its builder, the raw per-image predictions behind every table and figure, the perceptual-hash contamination lists, and the Kaggle notebooks that produced the segmentation retraining of Section 4.11 are available at `https://github.com/Just-Ryan/burn-severity-benchmark`. Trained weights exceed the repository host's per-file limit and are attached to the tagged release there: both segmentation models retrained on the source-grouped split, both originals trained on the collection's distribution-supplied split, and the classifier shipped in the mobile demonstration. We publish the contaminated models alongside the corrected ones deliberately, so that the leak of Section 4.11 can be reproduced as readily as its correction. The benchmark's original Swin-Tiny classifier weights were not archived at the time; rather than leave the question open we retrained all ten under the original per-seed recipe and re-scored the pipeline arm on the corrected localiser, and the notebook that did so is in the repository. The scripts that generate every figure are there as well, so a reader can regenerate the figures as well as the numbers. That repository includes `verify_paper_numbers.py`, a self-contained script that recomputes **205** published quantities from the committed raw data without a GPU, a dataset download or model weights, and reports each against the value printed here; all 205 pass. Two of the checks exist specifically to catch us overstating our own results, one confirming that the matched-architecture leakage contrast is the smaller $2.07 \to 0.55$ rather than $3.06 \to 0.55$, and one confirming that the internal interval-narrowing factor is 3.35 rather than the "four times" an earlier draft of this manuscript claimed. The repository also includes `skin_tone_probe.py`, which reproduces the ITA analysis of Section 4.10 from the test images and the released per-image predictions.

References

Begoña Acha, Carmen Serrano, José I. Acha, and Laura M. Roa. Segmentation and classification of burn images by color and texture information. *Journal of Biomedical Optics*, 10(3):034014, 2005. doi:10.1117/1.1921227.

Mohammed A. Al-masni, Dong-Hyun Kim, and Tae-Seong Kim. Multiple skin lesions diagnostics via integrated deep convolutional networks for segmentation and classification. *Computer Methods and Programs in Biomedicine*, 190:105351, 2020. doi:10.1016/j.cmpb.2020.105351.

Binus. Skin burn wound classification dataset. Roboflow Universe, 2023. URL `https://universe.roboflow.com/binus-if3z9/skin-burn-wound-classification`. Accessed 20 July 2026.

Constance Boissin, Lucie Laflamme, Jian Fransén, Mikael Lundin, Fredrik Huss, Lee Wallis, Nikki Allorto, and Johan Lundin. Development and evaluation of deep learning algorithms for assessment of acute burns and the need for surgery. *Scientific Reports*, 13:1794, 2023. doi:10.1038/s41598-023-28164-4.

Xavier Bouthillier, Pierre Delaunay, Mirko Bronzi, Assya Trofimov, Brennan Nichyporuk, Justin Szeto, Nazanin Mohammadi Sepahvand, Edward Raff, Kanika Madan, Vikram Voleti, Samira Ebrahimi Kahou, Vincent Michalski, Tal Arbel, Chris Pal, Gaël Varoquaux, and Pascal Vincent. Accounting for variance in machine learning benchmarks. In *Proceedings of Machine Learning and Systems*, volume 3, pages 747–769, 2021. URL `https://proceedings.mlsys.org/paper_files/paper/2021/hash/0184b0cd3cfb185989f858a1d9f5c1eb-Abstract.html`.

Pierangela Bruno, Megan Macrì, and Carmine Dodaro. A dual-stage deep learning framework for breast ultrasound image segmentation and classification. *Journal of Medical Systems*, 49(1):162, 2025. doi:10.1007/s10916-025-02298-6.

Nele Brusselaers, Stan Monstrey, Dirk Vogelaers, Eric Hoste, and Stijn Blot. Severe burn injury in Europe: a systematic review of the incidence, etiology, morbidity, and mortality. *Critical Care*, 14(5):R188, 2010. doi:10.1186/cc9300.

Che Wei Chang, Feipei Lai, Mesakh Christian, Yu Chun Chen, Ching Hsu, Yo Shen Chen, Dun Hao Chang, Tyng Luen Roan, et al. Deep learning-assisted burn wound diagnosis: diagnostic model development study. *JMIR Medical Informatics*, 9(12):e22798, 2021. doi:10.2196/22798.

Che Wei Chang, Chun Yee Ho, Feipei Lai, Mesakh Christian, Shih Chen Huang, Dun Hao Chang, and Yo Shen Chen. Application of multiple deep learning models for automatic burn wound assessment. *Burns*, 49(5):1039–1051, 2023. doi:10.1016/j.burns.2022.07.006.

A. Chardon, I. Cretois, and C. Hourseau. Skin colour typology and suntanning pathways. *International Journal of Cosmetic Science*, 13(4):191–208, 1991. doi:10.1111/j.1467-2494.1991.tb00561.x.

Justin Dang, Matthew Lin, Calvin Tan, Christopher H Pham, Samantha Huang, Ian F Hulsebos, Haig Yenikomshian, and Justin Gillenwater. Use of infrared thermography for assessment of burn depth and healing potential: a systematic review. *Journal of Burn Care and Research*, 42(6):1120–1127, 2021. doi:10.1093/jbcr/irab108.

Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, et al. ImageNet: a large-scale hierarchical image database. In *IEEE Conference on Computer Vision and Pattern Recognition*, pages 248–255, Miami, FL, USA, 2009.

Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, et al. An image is worth 16x16 words: transformers for image recognition at scale. In *International Conference on Learning Representations*, Vienna, Austria, 2021.

Ahmed Elsarta, Habiba Fathalla, Marina Nasser, Sara Elwatany, Rawan Fekry, Mohamed Ebaid, Yomna Mahmoud, Ayman Anwar, et al. Integrating multi-source data for skin burn classification using deep learning. *Computers in Biology and Medicine*, 195:110556, 2025. doi:10.1016/j.compbiomed.2025.110556.

Andre Esteva, Brett Kuprel, Roberto A. Novoa, Justin Ko, Susan M. Swetter, Helen M. Blau, and Sebastian Thrun. Dermatologist-level classification of skin cancer with deep neural networks. *Nature*, 542(7639):115–118, 2017. doi:10.1038/nature21056.

Julius Ferdinand, Davy Viriya Chow, and Simeon Yuda Prasetyo. Automated skin burn detection and severity classification using YOLO convolutional neural network pretrained model. *E3S Web of Conferences*, 426: 01076, 2023. doi:10.1051/e3sconf/202342601076.

Yu-Tung Fu. Convolutional neural networks for burn segmentation and classification tasks using RGB photographs: a five-year systematic review. *Burns Open*, 14:100452, 2026. doi:10.1016/j.burnso.2026.100452.

Davide Griffa, Alessio Natale, Yuri Merli, Michela Starace, Nico Curti, Martina Mussi, Gastone Castellani, Davide Melandri, et al. Artificial intelligence in wound care: a narrative review of the currently available mobile apps for automatic ulcer segmentation. *BioMedInformatics*, 4(4):2321–2337, 2024. doi:10.3390/biomedinformatics4040126.

Sami Hamdan, Bradley C Love, Georg G von Polier, Susanne Weis, Holger Schwender, Simon B Eickhoff, and Kaustubh R Patil. Confound-leakage: confound removal in machine learning leads to leakage. *GigaScience*, 12: giad071, 2023. doi:10.1093/gigascience/giad071.

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *IEEE Conference on Computer Vision and Pattern Recognition*, pages 770–778, Las Vegas, NV, USA, 2016.

David M. Heimbach, Martin A. Afromowitz, Loren H. Engrav, Janet A. Marvin, and Bernice Perry. Burn depth estimation: man or machine. *Journal of Trauma*, 24 (5):373–378, 1984. doi:10.1097/00005373-198405000-00001.

Andrew Howard, Mark Sandler, Grace Chu, Liang-Chieh Chen, Bo Chen, et al. Searching for MobileNetV3. In *IEEE/CVF International Conference on Computer Vision*, pages 1314–1324, Seoul, Korea, 2019.

Gao Huang, Zhuang Liu, Laurens van der Maaten, and Kilian Q. Weinberger. Densely connected convolutional networks. In *IEEE Conference on Computer Vision and Pattern Recognition*, pages 4700–4708, Honolulu, HI, USA, 2017.

Douglas MacG Jackson. The diagnosis of the depth of burning. *British Journal of Surgery*, 40(164):588–596, 1953. doi:10.1002/bjs.18004016413.

Amín D. Jaskille, Jeffrey W. Shupp, Marion H. Jordan, and James C. Jeng. Critical review of burn depth assessment techniques: part I, historical review. *Journal of Burn Care and Research*, 30(6):937–947, 2009. doi:10.1097/BCR.0b013e3181c07f21.

Glenn Jocher, Ayush Chaurasia, and Jing Qiu. Ultralytics YOLOv8. Software, 2023. URL `https://github.com/ultralytics/ultralytics`.

Sayash Kapoor and Arvind Narayanan. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9):100804, 2023. doi:10.1016/j.patter.2023.100804.

Sayash Kapoor, Emily M. Cantrell, Kenny Peng, Thanh Hien Pham, Christopher A. Bail, Odd Erik Gundersen, Jake M. Hofman, Jessica Hullman, et al. REFORMS: consensus-based recommendations for machine-learning-based science. *Science Advances*, 10(18):eadk3452, 2024. doi:10.1126/sciadv.adk3452.

Shachar Kaufman, Saharon Rosset, Claudia Perlich, and Ori Stitelman. Leakage in data mining: formulation, detection, and avoidance. *ACM Transactions on Knowledge Discovery from Data*, 6(4):1–21, 2012. doi:10.1145/2382577.2382579.

Jure Kovac, Peter Peer, and Franc Solina. Human skin colour clustering for face detection. In *IEEE Region 8 EUROCON 2003. Computer as a Tool*, volume 2, pages 144–148, 2003. doi:10.1109/EURCON.2003.1248169.

Chun Hon Lau, Ken Hung-On Yu, Tsz Fung Yip, Luke Yik Fung Luk, Abraham Ka Chung Wai, Tin-Yan Sit, Janet Yuen-Ha Wong, and Joshua Wing Kei Ho. An artificial intelligence-enabled smartphone app for real-time pressure injury assessment. *Frontiers in Medical Technology*, 4:905074, 2022. doi:10.3389/fmedt.2022.905074.

Geert Litjens, Thijs Kooi, Babak Ehteshami Bejnordi, Arnaud Arindra Adiyoso Setio, Francesco Ciompi, Mohsen Ghafoorian, Jeroen A.w.m. van der Laak, Bram van Ginneken, et al. A survey on deep learning in medical image analysis. *Medical Image Analysis*, 42:60–88, 2017. doi:10.1016/j.media.2017.07.005.

Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, et al. Swin transformer: hierarchical vision transformer using shifted windows. In *IEEE/CVF International Conference on Computer Vision*, pages 10012–10022, Montreal, Canada, 2021.

Zhuang Liu, Hanzi Mao, Chao-Yuan Wu, Christoph Feichtenhofer, Trevor Darrell, et al. A ConvNet for the 2020s. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 11976–11986, New Orleans, LA, USA, 2022.

Jun Ma, Yuting He, Feifei Li, Lin Han, Chenyu You, and Bo Wang. Segment anything in medical images. *Nature Communications*, 15(1), 2024. doi:10.1038/s41467-024-44824-z.

Lena Maier-Hein, Matthias Eisenmann, Annika Reinke, Sinan Onogur, Marko Stankovic, Patrick Scholz, Tal Arbel, Hrvoje Bogunovic, et al. Why rankings of biomedical image analysis competitions should be interpreted with care. *Nature Communications*, 9(1):5217, 2018. doi:10.1038/s41467-018-07619-7.

Lena Maier-Hein, Annika Reinke, Patrick Godau, Minu D. Tizabi, Florian Buettner, Evangelia Christodoulou, Ben Glocker, Fabian Isensee, et al. Metrics reloaded: recommendations for image analysis validation. *Nature Methods*, 21(2):195–212, 2024. doi:10.1038/s41592-023-02151-z.

Quinn McNemar. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157, 1947. doi:10.1007/BF02295996.

Stan Monstrey, Henk Hoeksema, Jos Verbelen, Ali Pirayesh, and Phillip Blondeel. Assessment of burn depth and burn wound healing potential. *Burns*, 34(6):761–769, 2008. doi:10.1016/j.burns.2008.01.009.

Gustav Mårtensson, Daniel Ferreira, Tobias Granberg, Lena Cavallin, Ketil Oppedal, Alessandro Padovani, Irena Rektorova, Laura Bonanni, et al. The reliability of a deep learning model in clinical out-of-distribution MRI data: a multicohort study. *Medical Image Analysis*, 66:101714, 2020. doi:10.1016/j.media.2020.101714.

Aaiza Nadeem, Bushra Bashir, Samman Khalid, Irum Matloob, Alishba Sajid, Syeda Minahil Zahra, Husnain Khan, Fatima Khalique, et al. A transformer-based computational framework for automated burn injury segmentation and severity assessment. *IEEE Access*, 14:27793–27811, 2026. doi:10.1109/ACCESS.2026.3663629.

Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, Mahmoud Assran, Nicolas Ballas, et al. DINOv2: Learning robust visual features without supervision. *arXiv preprint arXiv:2304.07193*, 2023. URL `https://arxiv.org/abs/2304.07193`.

Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, et al. PyTorch: an imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems*, pages 8024–8035, Vancouver, Canada, 2019.

Michael D. Peck. Epidemiology of burns throughout the world. Part I: distribution and risk factors. *Burns*, 37(7): 1087–1100, 2011. doi:10.1016/j.burns.2011.06.005.

David Picard. torch.manual_seed(3407) is all you need: on the influence of random seeds in deep learning architectures for computer vision. *arXiv preprint arXiv:2109.08203*, 2021. doi:10.48550/arXiv.2109.08203.

Saeka Rahman, Md Motiur Rahman, Miad Faezipour, Mo Rastgaar, Elika Ridelman, Justin D. Klein, Beth A. Angst, and Christina M. Shanti. Enhancing burn severity assessment with deep learning: a comparative analysis and computational efficiency evaluation. *IEEE Access*, 12:147249–147268, 2024. doi:10.1109/ACCESS.2024.3476110.

Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, Eric Mintun, Junting Pan, Kalyan Vasudev Alwala, Nicolas Carion, Chao-Yuan Wu, Ross Girshick, Piotr Dollár, and Christoph Feichtenhofer. SAM 2: Segment anything in images and videos. *arXiv preprint arXiv:2408.00714*, 2024. URL https://arxiv.org/abs/2408.00714.

Michael Roberts, Derek Driggs, Matthew Thorpe, Julian Gilbey, Michael Yeung, Stephan Ursprung, Angelica I. Aviles-Rivero, Christian Etmann, et al. Common pitfalls and recommendations for using machine learning to detect and prognosticate for COVID-19 using chest radiographs and CT scans. *Nature Machine Intelligence*, 3: 199–217, 2021. doi:10.1038/s42256-021-00307-0.

Matthew Rosenblatt, Link Tejavibulya, Rongtao Jiang, Stephanie Noble, and Dustin Scheinost. Data leakage inflates prediction performance in connectome-based machine learning models. *Nature Communications*, 15(1): 1829, 2024. doi:10.1038/s41467-024-46150-w.

Behrouz Rostami, Jeffrey Niezgoda, Sandeep Gopalakrishnan, and Zeyun Yu. Multiclass burn wound image classification using deep convolutional neural networks. *arXiv preprint arXiv:2103.01361*, 2021.

Théo Sourget, Michelle Hestbek-Møller, Amelia Jiménez-Sánchez, Jack Junchi Xu, and Veronika Cheplygina. Mask of truth: model sensitivity to unexpected regions of medical images. *Journal of Imaging Informatics in Medicine*, 39:767–784, 2025. doi:10.1007/s10278-025-01531-5.

Sayma Alam Suha and Tahsina Farah Sanam. A deep convolutional neural network-based approach for detecting burn severity from skin burn images. *Machine Learning with Applications*, 9:100371, 2022. doi:10.1016/j.mlwa.2022.100371.

Iulian Emil Tampu, Anders Eklund, and Neda Haj-Hosseini. Inflation of test accuracy due to data leakage in deep learning-based classification of OCT images. *Scientific Data*, 9(1):580, 2022. doi:10.1038/s41597-022-01618-6.

Mingxing Tan and Quoc V. Le. EfficientNet: rethinking model scaling for convolutional neural networks. In *International Conference on Machine Learning*, pages 6105–6114, Long Beach, CA, USA, 2019.

Jeffrey E Thatcher, Faliu Yi, Amy E Nussbaum, John Michael DiMaio, Jason Dwight, Kevin Plant, Jeffrey E Carter, and James H Holmes. Clinical investigation of a rapid non-invasive multispectral imaging device utilizing an artificial intelligence algorithm for improved burn assessment. *Journal of Burn Care and Research*, 44 (4):969–981, 2023. doi:10.1093/jbcr/irad051.

Gaël Varoquaux and Veronika Cheplygina. Machine learning for medical imaging: methodological failures and recommendations for the future. *npj Digital Medicine*, 5(1): 48, 2022. doi:10.1038/s41746-022-00592-y.

Ross Wightman. PyTorch image models. Software, 2019. URL https://github.com/huggingface/pytorch-image-models.

Edwin B. Wilson. Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158):209–212, 1927. doi:10.1080/01621459.1927.10502953.

Sanghyun Woo, Shoubhik Debnath, Ronghang Hu, Xinlei Chen, Zhuang Liu, In So Kweon, and Saining Xie. ConvNeXt V2: Co-designing and scaling convnets with masked autoencoders. *arXiv preprint arXiv:2301.00808*, 2023. URL https://arxiv.org/abs/2301.00808.

World Health Organization. Burns. Fact sheet, WHO, Geneva, Switzerland, 2023. URL https://www.who.int/news-room/fact-sheets/detail/burns.

D. P. Yadav, Turki Aljrees, Deepak Kumar, Ankit Kumar, Kamred Udham Singh, and Teekam Singh. Spatial attention-based residual network for human burn identification and classification. *Scientific Reports*, 13:12516, 2023. doi:10.1038/s41598-023-39618-0.

Ekin Yagis, Selamawet Workalemahu Atnafu, Alba García Seco de Herrera, Chiara Marzi, Riccardo Scheda, Marco Giannelli, Carlo Tessa, Luca Citi, et al. Effect of data leakage in brain MRI classification using 2d convolutional neural networks. *Scientific Reports*, 11(1):22544, 2021. doi:10.1038/s41598-021-01681-w.

Alice C. Yu, Bahram Mohajer, and John Eng. External validation of deep learning algorithms for radiologic diagnosis: a systematic review. *Radiology: Artificial Intelligence*, 4(3):e210064, 2022. doi:10.1148/ryai.210064.

Metin Yıldız, Yakup Sarpdağı, Mehmet Okuyar, Mehmet Yildiz, Necmettin Çiftci, Ayşe Elkoca, Mehmet Salih Yildirim, Muhammet Ali Aydin, et al. Segmentation and classification of skin burn images with artificial intelligence: development of a mobile application. *Burns*, 50 (4):966–979, 2024. doi:10.1016/j.burns.2024.01.007.

Christoph Zauner. Implementation and benchmarking of perceptual image hash functions. Master's thesis, Upper Austria University of Applied Sciences, Hagenberg, Austria, 2010.

John R. Zech, Marcus A. Badgeley, Manway Liu, Anthony B. Costa, Joseph J. Titano, and Eric Karl Oermann. Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: a cross-sectional study. *PLOS Medicine*, 15(11): e1002683, 2018. doi:10.1371/journal.pmed.1002683.